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外文资料翻译

毕业设计题目 基于 3D 高斯泼溅的视觉导航算
法与实现

外文资料题目 GaussNav: Gaussian Splatting for
Visual Navigation

学 院 计算机科学与技术学院

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GaussNav: 面向视觉导航的3D高斯泼溅技术

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摘要—在具身视觉领域, 实例图像目标导航 (Instance ImageGoal Navigation, IIN) 要求智能体在未知环境中定位目标图像所描绘的特定物体。IIN任务的核心挑战在于, 智能体需要在不同视角下识别目标物体, 同时忽略潜在的干扰物。现有的基于地图的导航方法通常使用鸟瞰图 (Bird's Eye View, BEV) 地图, 这类地图缺乏场景的精细纹理表示。因此, 尽管BEV地图在语义级视觉导航任务中表现出色, 但在实例级任务中却难以胜任。

为此, 我们提出了一种全新的IIN任务框架——面向视觉导航的3D高斯泼溅技术 (Gaussian Splatting for Visual Navigation, GaussNav), 该框架基于3D高斯泼溅 (3D Gaussian Splatting, 3DGS) 构建了一种新颖的地图表示形式。GaussNav框架使智能体能够同时记忆场景的几何信息和语义信息, 并保留物体的纹理特征。

通过将相似物体的渲染结果与目标图像进行匹配, 智能体可以准确地识别、定位并导航至指定物体。我们的GaussNav框架实现了显著的性能提升, 在极具挑战性的Habitat-Matterport 3D (HM3D) 数据集上, 路径长度加权成功率 (Success weighted by Path Length, SPL) 从0.347提升至0.578。源代码已公开发布于: <https://github.com/XiaohanLei/GaussNav>。

Index Terms—具身视觉导航, 3D高斯泼溅。

I. 引言

身视觉导航是一个新兴的计算机视觉问题, 它要求具智能体利用视觉感知主动与世界交互并完成导航任务 [1]–[12]。近年来, 随着大规模真实感三维场景数据集 [13]–[15]和快速具身导航模拟器 [16]–[18]的出现, 具身视觉导航领域取得了长足进步。这些进展使研究人员能够在高度模拟真实世界条件的受控环境中开发和测试导航算法。

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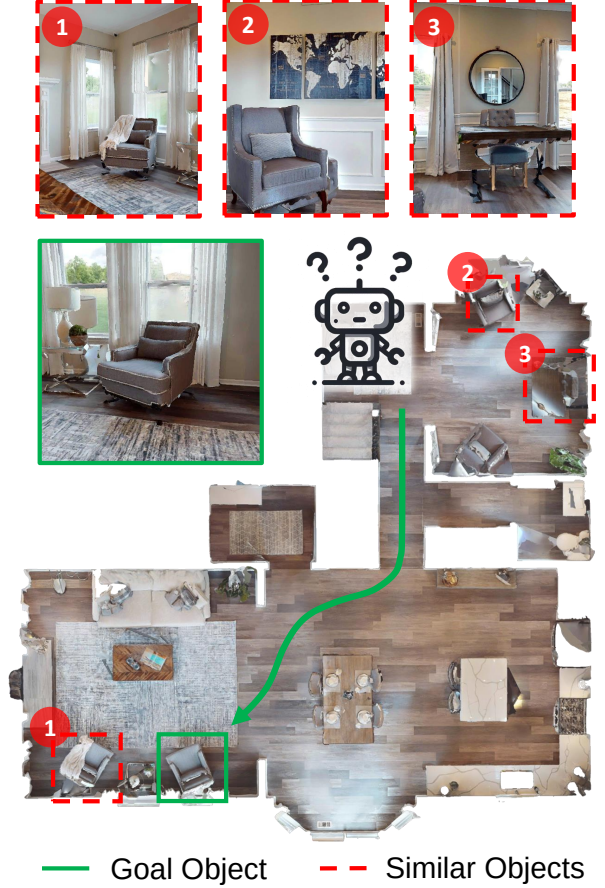


图 1. 实例图像目标导航(IIN)示意图。该任务要求智能体导航至目标图像所描绘的物体实例, 同时将其与其他视觉相似的实例区分开来。

在具身视觉导航中, 一个核心问题是“我们如何描述目标对象”。一种常见的任务设定是向智能体提供自然语言指令, 例如“检查我的笔记本电脑是否在椅子上”。然而, 当房间里有多把椅子时, 这种设定就会产生歧义。为解决这一挑战, Krantz等人 [19]提出了实例图像目标导航(IIN) [20]–[22]。在IIN任务中, 智能体被提供一张特定物体的图像, 其目标是在最短时间预算内导航至该特定物体。如Figure 1所示, 目标图像无需与导航智能体的传感器规格或身体形态相匹配。为完成该任务, 智能体需要在不同视角下识别目标物体, 并忽略潜在的干扰物。这是一项极具挑战性的任务, 因为它涉及语义推理、几何理解和实例感知匹配。

为解决上述问题, 现有方法 [20], [22]–[27]引入了二维语义鸟瞰图(BEV)地图来解决该问题。这些精心设计的显式地图表示具有内存高效的特点, 能够存储二维几何和语义等关键信息, 并可直接用于计算智能体的后续动作。然而, 这种简单的二维BEV地图表示缺乏保留环境三维几何信息的能力, 因此无法有效处理跨楼层导航场景。此外, BEV地图无法保留场景中的实例感知特征, 而这些特征对于区分同一类别的多个物体或需要细粒度物体交互的任务至关重要。

为克服BEV地图的局限性, 我们提出了一种全新的高斯泼溅导航框架, 即GaussNav, 用于解决IIN任务。我们的GaussNav框架受到三维视觉技术最新进展的启发, 包括神经辐射场(NeRF) [28]和三维高斯泼溅(3DGS) [29]。这些技术在新视角合成(NVS)和三维场景理解方面取得了重大突破。尽管这些技术最初并非为导航任务开发, 但它们在领域具有巨大的应用潜力。为此, 我们提出了语义高斯地图表示, 它整合了几何、语义和实例感知特征的表示, 可直接用于视觉导航。

我们的GaussNav框架包含三个阶段: 前沿探索、语义高斯构建和高斯导航。首先, 智能体通过前沿探索收集未知环境的观测数据。其次, 利用收集到的观测数据构建语义高斯。通过语义分割算法 [30], [31], 我们为每个高斯分配语义标签。然后根据语义标签和三维位置对高斯进行聚类, 将场景中的物体分割为不同语义类别下的不同实例。这种表示不仅能够保留场景的三维几何和每个高斯的语义标签, 还能保留场景的纹理细节, 从而支持新视角合成。最后, 我们为物体实例渲染描述性图像, 将其与目标图像进行匹配, 从而有效定位目标物体。在确定预测目标物体的位置后, 我们可以高效地将语义高斯转换为栅格地图, 并使用路径规划算法完成导航。

据我们所知, 我们是首次将三维高斯泼溅(3DGS) [29]引入具身视觉导航领域的研究者。在本工作中, 我们统一了视觉导航中几何、语义和实例感知特征的地图表示。我们的框架能够通过单张目标图像输入直接定位目标物体, 并引导智能体导航至目标, 无需额外的探索或验证步骤 [22]。我们的框架设计有助于实现高效且有效的视觉导航。我们在有效性和效率两个方面对所提方法进行了评估, 并在极具挑战性的Habitat-Matterport 3D数据集(HM3D) [14]上创下了新的最优性能记录。

II. 相关工作

我们首先简要讨论可微渲染领域的相关研究, 然后概述具身视觉导航的整体发展, 最后重点介绍与我们工作最相关的实例图像目标导航(IIN)任务。

可微渲染。 为实现真实感场景重建, 随着神经辐射场(NeRF) [28]的提出, 可微体渲染技术得到了广泛关注。NeRF使用单个多层感知机(MLP)表示场景, 通过沿像素光线行进并查询MLP获取不透明度和颜色来执行体渲染。由于体渲染本身具有可微性, MLP表示可以利用多视图信息通过最小化渲染损失进行优化, 从而实现高质量的新视角合成(NVS)。NeRF的主要局限性在于训练速度较慢。近期的研究通过引入显式体结构(如多分辨率体素网格 [32]–[34]和哈希函数 [35])来提升性能, 解决了这一问题。

与NeRF不同, 三维高斯泼溅(3DGS) [29]采用可微光栅化技术。与沿像素光线迭代的光线行进不同, 3DGS对待光栅化的图元进行迭代, 这与传统图形学光栅化方式类似。通过利用三维场景的天然稀疏性, 3DGS实现了既能捕捉高保真三维场景又能显著提升渲染速度的表达能力。关于三维高斯泼溅技术发展的全面综述 [36]强调了该方法的通用性及其在各个领域的应用。凭借这些优势, 越来越多的研究开始探索各种创新方向, 包括可变形或动态高斯 [37]–[39]、网格提取和物理模拟的进展 [40]–[42], 以及在同时定位与地图构建(SLAM)中的应用 [43]–[45]。对3DGS能力的广泛关注促使我们思考其在具身视觉导航中的潜力。鉴于高斯表示天然包含显式的场景几何信息和渲染所需的参数, 我们认为3DGS能够增强基于地图的视觉导航中的决策能力。高斯表示的显式特性提供了丰富且紧凑的环境数据形式, 可有效用于指导自主智能体的导航行为。

具身视觉导航。 具身视觉导航包含多个研究方向: 物体目标导航(ObjectNav)、多物体目标导航(MultiON)、图像目标导航(ImageNav)和实例图像目标导航(IIN)。物体目标导航 [24], [46]–[49]要求智能体导航至环境中指定物体类别的任意实例。多物体目标导航 [50]–[53]则要求智能体依次导航至一系列物体。图像目标导航 [26], [54]–[61]涉及导航至拍摄目标图像的相机位姿。相比之下, 实例图像目标导航 [20]–[22]要求导航至目标图像中相机所拍摄的特定实例。总体而言, 这些导航任务涵盖了从ObjectNav和MultiON的语义级导航到ImageNav和IIN的细粒度实例级导航的完整范围, 全

面覆盖了具身视觉导航的问题空间。

许多解决具身视觉导航的方法使用深度强化学习(DRL)开发端到端策略,将以自我为中心的视觉信息映射为动作 [26], [46], [56], [62]。然而,在端到端框架中获取视觉场景理解、语义探索和长期记忆等技能具有挑战性。因此,这些方法通常结合精心设计的奖励塑造 [54]、预训练流程 [56]和先进的记忆模块 [59], [63], [64]。与端到端DRL不同,另一种方法将问题分解为可以以监督方式优化的子任务。这些子任务包括通过拓扑SLAM进行图预测 [58]、基于图的距离学习 [65], [66]以及用于最后一公里导航的相机位姿估计 [61]。Chaplot等人 [24]将具身视觉导航任务分解为探索、物体检测和局部导航。在此基础上,“车轮上的CLIP”(CoW) [67]采用了类似的分解策略,专注于探索和物体定位以处理开放集物体词汇。模块化方法在有效的仿真到现实迁移(Sim2Real)方面也显示出潜力。Gervet等人 [68]对模块化系统和端到端系统进行了Sim2Real迁移实验,证明模块化能够有效缓解端到端策略性能的视觉Sim2Real差距。

模块化方法通常使用地图表示环境,并利用该地图获取在该环境中导航所需的知识。鸟瞰图(BEV)地图 [20], [22], [24], [69]是一种度量地图,它从俯视角度投影整个场景,表示特定区域的占用状态、探索状态和语义类别。这种地图表示将每个区域的信息编码到网格中,实现了精确的空间感知。相比之下,拓扑地图 [57], [58], [60]更侧重于描述场景中不同区域之间的空间关系。这种方法允许智能体基于空间的连通性进行导航,而不仅仅依赖几何坐标。为了更精细地表示场景,研究人员提出了三维感知地图 [49], [70]–[72]。引入三维一致性可以增强导航中的感知能力。在三维地图的基础上,我们进一步采用三维高斯泼溅(3DGS) [29]作为一种新颖的地图表示。这使得地图能够合成特定物体实例的外观视图,我们已证明这在实例图像目标导航任务中是有效的。

实例图像目标导航。与图像目标导航相比,IIN任务带来了独特的挑战。首先,在IIN中,目标图像必须描绘一个特定的物体实例。而图像目标导航可以使用随机拍摄的照片,这些照片可能包含无关紧要的元素,例如大面积的白墙。其次,拍摄目标图像的相机参数不一定与智能体的相机参数匹配。因此,要在IIN任务中取得成功,智能体必须能够在众多与目标物体同属一类的候选物体中识别出目标物体,并能从各种视角识

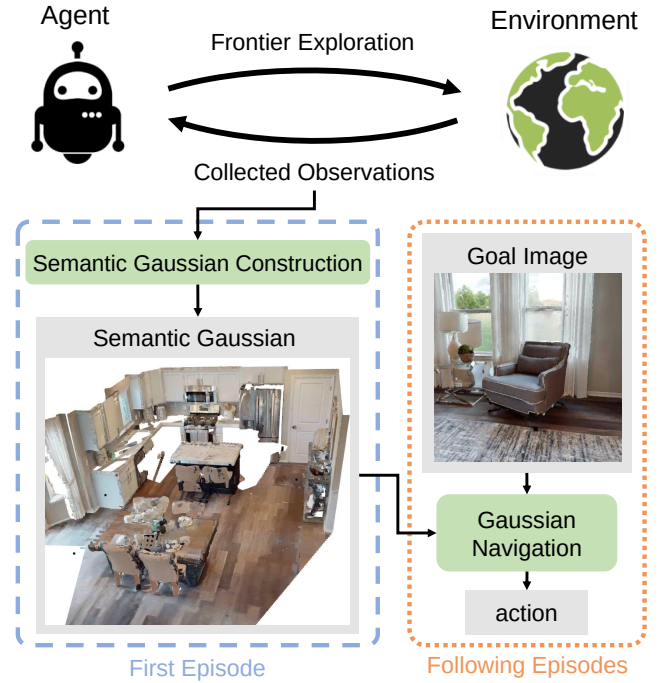


图 2. 框架总览。在场景的第一个回合中,智能体使用前沿探索收集未知环境的观测数据,构建语义高斯地图。在后续回合中,高斯导航模块利用预构建的语义高斯地图定位目标物体并引导智能体向其移动。

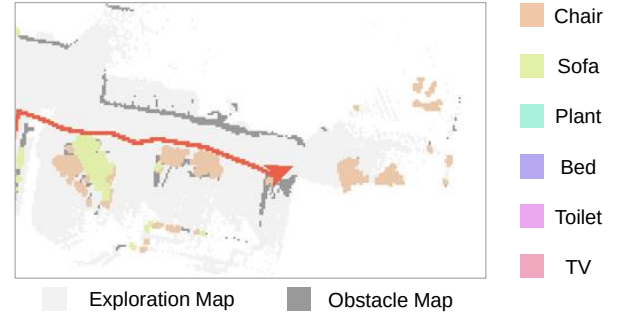


图 3. 探索地图与障碍物地图。

别它。为解决上述挑战,Krantz等人 [20]开发了一个通用流水线,用于对齐不同视角下的同一物体。Bono等人 [21]提出了一种端到端的IIN任务方法,而Lei等人 [22]提出了一种模仿人类远距离验证物体行为的方法。现有方法部分侧重于设计复杂的模块 [20], [22],部分侧重于在前置任务上进行预训练 [21]。我们的方法不同之处在于,我们专注于设计一种新的地图表示。通过这种新颖的地图形式,我们能够更好地建立目标描述与目标位置之间的联系,从而促进视觉导航。

III. 方法

A. 概述

在实例图像目标导航(IIN)任务中,新回合 e 开始时,智能体将获得一张包含指定物体实例 O_g 的目标图像 I_g 。

智能体的目标是在环境中导航至该目标实例 O_g 。在每个时间步 t ，智能体获取的观测数据包括RGB图像 I_t 、深度图像 D_t 和传感器位姿读数 P_t 。智能体必须利用这些信息决定并执行动作 a_t 。当智能体调用停止动作时，只有当它处于目标物体的一定范围内，该回合才被判定为成功。

为完成IIN任务，我们提出了一个模块化框架——面向视觉导航的三维高斯泼溅技术(GaussNav)，如Figure 2所示。在新环境中，我们提出的框架将实例图像目标导航任务分解为三个阶段：前沿探索、语义高斯构建（如Figure 4所示）和高斯导航（如Figure 5所示）。首先，在未知环境的第一个回合中，智能体采用前沿探索策略探索环境并收集观测数据。随后，我们使用提出的语义高斯地图重建场景。在后续回合中，智能体利用语义高斯地图定位目标图像中描绘的物体实例 o 并导航至该位置。这一过程有效地将IIN任务转化为更易处理的点目标导航(PointGoal Navigation)任务。

B. 前沿探索

在未探索环境的第一个回合中，智能体同时维护两种地图：探索地图和障碍物地图，如Figure 3所示。探索地图描绘了环境中已被探索的区域，而障碍物地图则标记了场景中的障碍物。通过检测探索地图的轮廓并排除障碍物地图中的区域，智能体将最近的前沿点设置为航点以推进探索。基于前沿的探索策略是机器人学和自主导航领域中一种成熟的方法 [73], [74]。它涉及识别环境中已探索区域与未探索区域之间的边界，即前沿。然后，智能体选择最近的可达前沿点作为下一个探索目标。这一决策基于到前沿的距离、路径的可通行性以及探索该前沿可能获得的信息增益 [75]。通过迭代探索最近的前沿点，智能体能够高效地覆盖整个环境，同时避开障碍物和已探索区域。通过应用这种基于前沿的探索策略，智能体收集到了整个环境的观测数据。

C. 语义高斯构建

我们的语义高斯地图由一组高斯表示，每个高斯由最少9个参数表征：RGB颜色向量三元组 $\mathbf{c}$ 、质心坐标三元组 $\boldsymbol{\mu} \in \mathbb{R}^3$ 、半径标量 r 、不透明度标量 $o \in [0, 1]$ 以及类别标签标量 l 。与原始三维高斯泼溅(3DGS) [29]不同，我们简化了高斯的表示，仅使用视角无关的颜色并将高斯约束为各向同性。这种简化在降低内存需求的同

时提高了计算效率。为全面理解三维高斯泼溅技术，我们强烈建议读者参考原始论文 [29]。我们的语义高斯构建过程可描述为一个包含两个交替步骤的迭代过程：高斯稠密化和语义高斯更新，如Figure 4所示。高斯稠密化在每个新帧到来时向语义高斯地图中添加新的高斯，而语义高斯更新则通过可微渲染优化每个高斯的参数。

可微渲染。 三维高斯泼溅(3DGS) [29]按如下方式渲染RGB图像：给定一组三维高斯和相机位姿，首先将所有高斯按从前到后的顺序排序。然后，通过在像素空间中按顺序对每个高斯的二维投影进行alpha合成，可以高效地渲染出RGB图像。像素 $\mathbf{p} = (u, v)$ 的渲染颜色可表示为：

$$\hat{I}(\mathbf{p}) = \sum_{i=1}^n \mathbf{c}_i f_i(\mathbf{p}) \prod_{j=1}^{i-1} (1 - f_j(\mathbf{p})), \quad (1)$$

其中 $f_i(\mathbf{p})$ 的计算方式如下：

$$f(\mathbf{x}) = o \exp\left(-\frac{\|\mathbf{x} - \boldsymbol{\mu}\|^2}{2r^2}\right). \quad (2)$$

$\boldsymbol{\mu}$ 和 r 是像素空间中投影后的二维高斯参数：

$$\boldsymbol{\mu}^{2D} = K \frac{E_t \boldsymbol{\mu}}{d}, \quad r^{2D} = \frac{fr}{d}, \quad \text{其中 } d = (E_t \boldsymbol{\mu})_z. \quad (3)$$

这里， K 是相机内参矩阵， E_t 是编码 t 时刻相机旋转和平移的外参矩阵， f 是已知的焦距， d 是第 i 个高斯相对于相机坐标系的深度。

渲染。 与三维高斯泼溅(3DGS) [29]不同，我们可以微地渲染深度图和轮廓图，这决定了可见性，并将用于后续的高斯稠密化和更新。像素 $\mathbf{p}$ 处的深度 $\hat{D}$ 和轮廓 $\hat{S}$ 的渲染方式如下：

$$\hat{D}(\mathbf{p}) = \sum_{i=1}^n d_i f_i(\mathbf{p}) \prod_{j=1}^{i-1} (1 - f_j(\mathbf{p})), \quad (4)$$

$$\hat{S}(\mathbf{p}) = \sum_{i=1}^n f_i(\mathbf{p}) \prod_{j=1}^{i-1} (1 - f_j(\mathbf{p})). \quad (5)$$

在此基础上，我们还渲染语义分割结果：

$$\hat{C}(\mathbf{p}) = \sum_{i=1}^n l_i f_i(\mathbf{p}) \prod_{j=1}^{i-1} (1 - f_j(\mathbf{p})). \quad (6)$$

语义分割。 如Figure 4所示，智能体的RGB观测 I_t 通过Mask-RCNN [30]被分割为 C_t 。分割结果将用于在高斯稠密化阶段初始化新的高斯，并在语义高斯更新阶段监督高斯参数的优化。

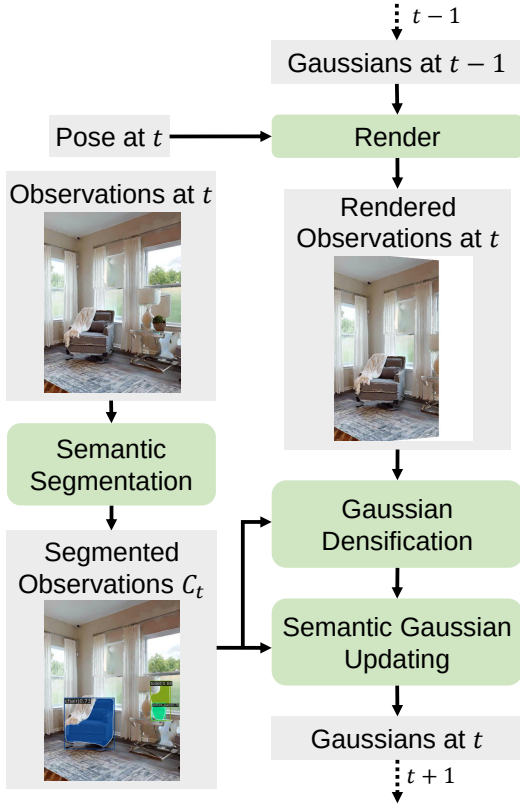


图 4. 语义高斯构建示意图。在时间步 t ，流水线通过稠密化和更新步骤优化 $t-1$ 时刻的高斯，这涉及将渲染的RGB和深度图像与当前输入的训练视图进行比较。同时，使用分割图像为稠密化后的高斯分配语义标签。最后，通过可微渲染对高斯进行精调。

高斯稠密化。 高斯稠密化通过将 $t-1$ 时刻的高斯在位置 P_i 处的渲染结果与真实值进行比较来执行。该过程在先前的高斯无法表示新观测中的场景区域时添加新的高斯。遵循Keetha等人 [43]的方法，我们基于稠密化掩码确定哪些像素需要进行稠密化：

$$M(\mathbf{p}) = \left(\hat{S}(\mathbf{p}) < 0.5 \right) + \left(D(\mathbf{p}) < \hat{D}(\mathbf{p}) \right) \left(L_1(\hat{D}(\mathbf{p})) > 50\text{MDE} \right), \quad (7)$$

其中第一项表示语义高斯不够稠密的区域，第二项表示真实深度在预测深度之前且深度误差大于50倍中值深度误差(MDE)的区域。

语义高斯更新。 在对当前语义高斯地图进行稠密化后，我们根据位姿和观测数据更新高斯的参数。这通过可微渲染和基于梯度的优化来实现，等价于将辐射场拟合到已知位姿图像的“经典”问题。具体而言，我们通过最小化RGB、深度和分割误差来更新高斯的参数。

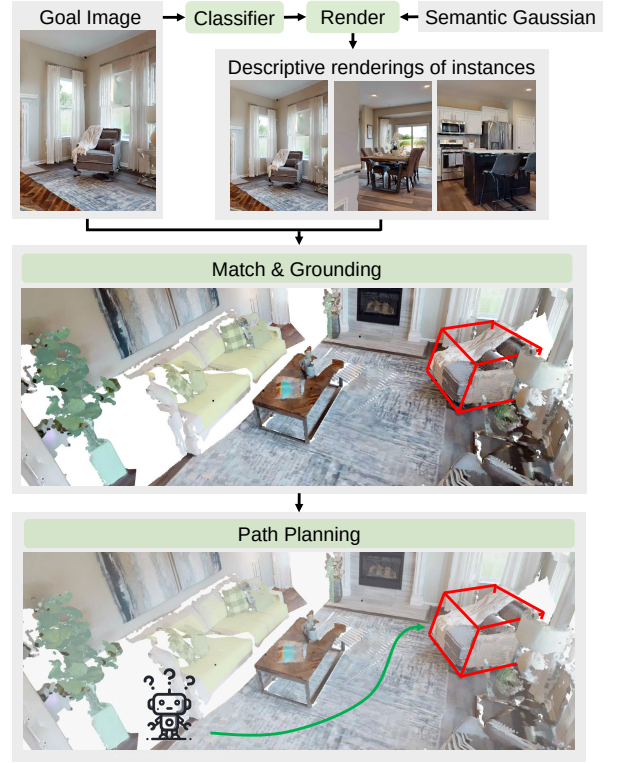


图 5. 高斯导航示意图。我们的方法首先使用预构建的语义高斯地图对目标图像进行分类。在确定预测类别后，我们围绕属于该类别的实例生成描述性图像。然后将这些图像与目标物体进行匹配，以识别并定位目标实例。利用地图和确定的目标，智能体采用路径规划算法计算动作序列。

D. 高斯导航

为利用构建好的语义高斯地图进行导航，我们提出了高斯导航方法，如Figure 5所示。我们首先对目标图像 I_g 进行分类，预测当前目标的语义标签 $\hat{l}_g$ ，例如Figure 5中的“椅子”。然后使用该语义标签查询相关的高斯。对于每个属于同一类别的相关物体实例，我们渲染描述性图像。将这些渲染结果与目标图像进行比较以定位目标物体，预测其位置 $\hat{P}_g$ 。最后，路径规划模块生成可行路径并确定智能体的动作。

分类器。 在IIN任务中，智能体接收一张描绘目标物体的图像 I_g 。然而，由于搜索空间巨大，将该图像与整个场景中所有可导航点的渲染结果进行比较会极其耗时。因此，借助我们的语义高斯地图 G_s ，我们只需搜索与目标物体对应类别的实例。为此，我们首先将目标图像 I_g 分类为目标类别标签 $\hat{l}_g$ 。我们使用HM3D-SEM [14]训练集上的目标图像对在ImageNet [76]上预训练的图像分类模型ResNet50 [77]进行微调。

匹配与定位。 利用预测的目标物体标签 $\hat{l}_g$ ，我们识别出所有具有相同类别标签的候选物体。对于每个候选实例，我们通过从多个视角渲染物体来生成一组描

述性图像，以捕捉其特征。具体而言，对于包含潜在候选物体的一个训练视图，我们通过新视角合成(NVS)将其增强为 n_v 个视图。我们将训练视图的相机到世界坐标系的变换记为 $c2w$ ，并将潜在目标物体在相机坐标系中的平移向量记为 t_c^o 。我们使用前向、右向和上向量定义物体到世界坐标系的旋转矩阵。前向方向是从训练视图到潜在目标物体的向量，即 t_c^o 。上向方向是向量 $[0, -1, 0]$ ，右向量与前向和上向量都正交，与右向、上向和前向量共同构成右手坐标系。物体到世界坐标系的平移向量为：

$$t_w^o = c2w \times t_c^o. \quad (8)$$

旋转矩阵和平移向量共同构成了从物体到世界坐标系的刚性变换矩阵 $o2w$ 。

我们将绕 y 轴和 x 轴的旋转矩阵分别定义为 $R_y(\theta)$ 和 $R_x(\theta)$ ，分别表示在水平和垂直方向上绕物体旋转 θ 角形成的新视角。因此，水平方向上新视角的相机到世界坐标系变换为：

$$c2w_h(\theta) = o2w \times R_y(\theta) \times w2o \times c2w, \quad (9)$$

垂直方向上为：

$$c2w_v(\theta) = o2w \times R_x(\theta) \times w2o \times c2w. \quad (10)$$

在实验中，当 $n_v = 1$ 时，我们不执行新视角合成；当 $n_v = 3$ 时，我们在 $\theta = \pm 15^\circ$ （水平和垂直方向）执行新视角合成；当 $n_v = 5$ 时，我们使用 $\theta = \pm 15^\circ$ 和 $\pm 30^\circ$ 。

新视角合成后，原始训练视图得到增强。第 i 个物体实例的增强渲染集记为 S_i 。设 n 为与目标物体同类别观测到的实例数，则 S 的全集可表示为：

$$S = \{S_1, S_2, \dots, S_n\} \quad \text{其中} \quad i = 1, 2, \dots, n. \quad (11)$$

为从这些候选物体中区分出目标物体，我们将问题表述为：

$$i_{\max} = \arg \max \left\{ \max_{s \in S_1} \Omega(s), \max_{s \in S_2} \Omega(s), \dots, \max_{s \in S_n} \Omega(s) \right\} \\ \text{其中} \quad i = 1, 2, \dots, n, \quad (12)$$

其中 $\Omega(\cdot)$ 定义为渲染图与目标图像 I_g 之间匹配的关键点数量。具体而言，对于第 i 个物体实例的渲染图 $s \in S_i$ 和目标图像 I_g ，我们使用DISK [78]提取关键点的像素坐标 (x, y) 及其关联的特征描述子 V_t 。即：

$$(K_t, V_t) = \text{DISK}(s), \quad (K_g, V_g) = \text{DISK}(I_g). \quad (13)$$

随后，使用LightGlue [79]计算匹配对 $(\hat{K}_t, \hat{K}_g)$ 。特征匹配过程可表述为：

$$(\hat{K}_t, \hat{K}_g) = \text{LightGlue}((K_t, V_t), (K_g, V_g)). \quad (14)$$

因此，匹配点的数量，即 $\hat{K}_t$ 或 $\hat{K}_g$ 的长度，记为 Ω 。渲染图像产生最多匹配关键点候选物体被选为目标物体，如Equation (12)所示。

当物体实例被选中后，我们在语义高斯地图中定位该物体。由于语义分割误差会导致离群点的存在，我们对地图上的实例进行聚类。具体而言，我们使用基于密度的空间聚类算法(DBSCAN)¹，该算法将紧密聚集的点归为一类，将低密度区域中孤立的点标记为离群点。通过精确确定所选物体实例的位置，我们可以轻松地将IIN任务转化为点目标任务。

路径规划。 语义高斯地图并非为路径规划设计。我们首先将语义高斯地图转换为点云，其中每个高斯被简化为点云表示中的一个点。然后将点云体素化为三维体素 M_{3D} ，再将三维体素 M_{3D} 投影为二维鸟瞰图栅格 M_{2D} 。这里，我们使用语义高斯地图的二维投影而非预探索阶段使用的二维几何地图，以保持整个流水线表示的一致性。我们的语义高斯地图使用从深度图像导出的点云初始化高斯，并且在优化阶段不修剪任何高斯。因此，语义高斯地图的二维投影与二维几何地图在本质上是等价的。

给定二维鸟瞰图栅格地图 M_{2D} ，以及智能体的起始位置和目标位置，我们可以使用快速行进法(FMM)高效生成最短距离场。该场中的每个点都包含从起点到该点所需的最小距离。我们提取该距离场中落在智能体操作范围内的相关子集。随后，从该子集中选择一个航点，确保它不与任何障碍物相交且位于距离场的局部最小值处。根据所选航点，智能体可以根据当前状态的角度和距离轻松计算出动作。智能体迭代执行上述过程生成动作序列，直到到达目的地。

IV. 实验

本节首先详细介绍我们的实验设置，然后与当前最优方法进行对比分析。随后，我们通过消融实验评估所提方法中各个组件的有效性。最后，我们评估了该方法的计算效率，分析了错误案例，并对渲染质量进行了评价。

¹<https://scikit-learn.org/stable/modules/generated/sklearn.cluster.DBSCAN.html>

Method	Success $\uparrow$	SPL $\uparrow$
RL Baseline [19]	0.083	0.035
OVRL-v2 ImageNav [55]	0.006	0.002
OVRL-v2 IIN [55]	0.248	0.118
FGPrompt [80]	0.099	0.028
MultiON Baseline [51]	0.066	0.045
MultiON Implicit [53]	0.143	0.107
MultiON Camera [50]	0.186	0.142
Mod-IIN [20]	0.561	0.233
IEVE Mask RCNN [22]	0.684	0.241
IEVE InternImage [22]	0.702	0.252
Mod-IIN [20] (Scene Map)	0.563	0.323
IEVE Mask RCNN [22] (Scene Map)	0.683	0.331
IEVE InternImage [22] (Scene Map)	0.705	0.347
GaussNav (ours)	0.725	0.578

表 1

PERFORMANCE COMPARISON OF OUR GAUSSNAV WITH PREVIOUS STATE-OF-THE-ART METHODS ON THE HM3D [14] DATASETS ACROSS TWO DIFFERENT METRICS: SUCCESS AND SPL [81]. THE TABLE IS DIVIDED INTO FOUR SECTIONS. THE FIRST SECTION PRESENTS THE RESULTS OF END-TO-END METHODS. THE SECOND SECTION SHOWS THE TRANSFER PERFORMANCE OF MULTION-RELATED METHODS ON THE IIN TASK. THE THIRD SECTION INCLUDES THE STATE-OF-THE-ART METHODS ON THE IIN TASK. FINALLY, THE FOURTH SECTION AIMS TO PROVIDE A FAIR COMPARISON WITH GAUSSNAV BY REPLACING THE EPISODIC MAP USED IN THESE METHODS FROM THE THIRD SECTION WITH A SCENE-SPECIFIC MAP, ALLOWING THE AGENT TO RETAIN THE MAP FROM THE PREVIOUS EPISODE.

A. 实验设置

数据集。 我们在Habitat仿真器 [16]中使用Habitat-Matterport 3D数据集(HM3D) [14]进行实验。HM3D由真实世界环境的三维重建场景组成，并带有语义标注。这些场景被划分为训练、验证和测试三个独立子集，分别包含145/36/35个场景。我们遵循Krantz等人 [19]提出的实例图像目标导航(IIN)任务设置。回合数据集同样被划分为训练、验证和测试三个子集，分别包含7056K/1K/1K个回合。目标图像所描绘的物体属于以下六个类别：{“椅子”，“沙发”，“植物”，“床”，“马桶”，“电视”}。在验证子集上，共观测到795个独特的物体实例。

智能体形态。 我们采用Hello Robot Stretch机器人平台²的形态参数。仿真智能体被建模为一个零转弯半径的刚体圆柱体，高度为1.41米，半径为0.17米。一个前向的RGB-D相机安装在1.31米的高度处。在每个时间步 t ，智能体的观测包括以自我为中心的RGB图像、深

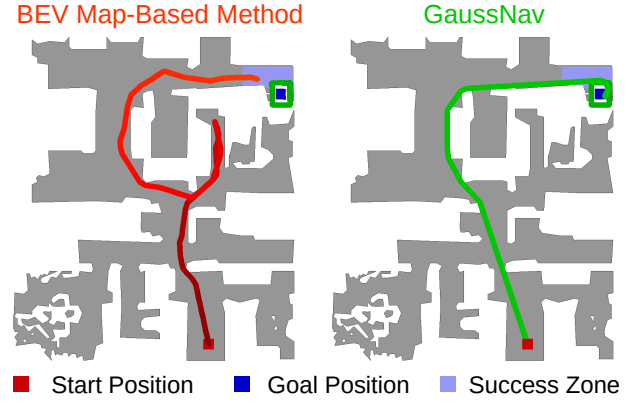


图 6. 轨迹分析。我们的语义高斯地图表示使智能体能够从单张目标图像直接定位目标物体，从而实现高效的视觉导航。

度图像、目标图像和传感器位姿读数。智能体相机与目标相机的规格（如安装高度、视角和视场角(FOV)）不同。具体而言，智能体相机的分辨率为 640×480 ，而目标相机的分辨率为 512×512 ，且高度和视场角参数不固定。

动作空间。 我们使用离散动作空间进行导航，包含四个动作：{停止(STOP)，前进(FORWARD)，右转(TURN_RIGHT)，左转(TURN_LEFT)}。停止动作终止当前回合，前进动作使智能体向前移动25厘米。旋转动作在原地执行：右转产生25度顺时针旋转，左转产生25度逆时针旋转。

评估指标。 遵循Krantz等人 [19]的方法，我们从成功率和效率两个方面评估模型。我们报告成功率(Success)和归一化逆路径长度加权成功率(SPL)。如果智能体在距离目标物体1.0米的欧氏距离内调用停止动作，则该回合被判定为成功 (Success = 1)。SPL是文献 [81]中定义的效率指标，计算公式为：

$$\text{SPL} = \frac{1}{N} \sum_{i=1}^N S_i \cdot \frac{l_i}{\max(p_i, l_i)}, \quad (15)$$

其中 N 是总回合数， S_i 是第 i 个回合的二元成功指示符， l_i 是从起始位置到目标的最短路径距离， p_i 是智能体实际走过的路径长度。SPL值越高表示导航效率越高。

B. 与当前最优方法的对比

我们将所提模型与各种基线方法和先前的最优工作进行了评估对比，结果如Table 1所示。与传统的IIN方法为每个回合生成一张地图不同，GaussNav在一个场景的多个回合之间构建统一的地图。第一部分列出了端到端基线方法的结果，第二部分包括为MultiON任务设计但在此处用于IIN任务的方法。

²<https://hello-robot.com/stretch-2>

Table I的第三部分展示了原始IIN最优方法的性能，第四部分报告了这些方法在适配使用场景地图后的性能。

端到端基线方法。 我们评估了两种端到端方法在IIN任务上的性能。*RL*基线构建了一个网络，输入包括智能体RGB图像($\mathcal{V}_{RGB}$)、智能体深度图像($\mathcal{V}_D$)、目标图像($\mathcal{V}_G$)、GPS坐标(x, z)和航向角(θ)。视觉观测使用独立的ResNet-18 [82]编码器进行编码，GPS和航向角使用32维线性层进行编码。然后将这些特征拼接起来，通过一个2层LSTM进行编码。最后，从类别分布中采样动作 $a(t)$ 。该网络使用近端策略优化(PPO) [83]从头开始训练。

Yadav等人 [55]提出的OVRL-v2为图像目标导航中的视觉编码器引入了自监督预训练。OVRL-v2最初是为图像目标导航任务训练的。直接将OVRL-v2应用于IIN任务而不进行微调会导致性能不佳，成功率仅为0.006 (Table I第2行)。这种性能下降可归因于多个因素：场景数据集从Gibson [13]切换到HM3D [14]、机器人形态从Locobot³变为Stretch，以及目标目的地的性质从图像来源变为图像主体。在HM3D数据集上针对IIN任务专门微调OVRL-v2后，性能显著提升，成功率达到0.248 (Table I第3行)。

MultiON任务中的最优方法。 MultiON任务与我们评估GaussNav时使用的场景特定地图表示有许多相似之处。我们在IIN基准上重新实现了几种MultiON任务的最优方法 [50], [51], [53]。值得注意的是，我们直接将目标物体的语义类别输入到上述方法中。由于IIN任务的导航要求是实例级而非语义级的，这种任务表述上的差异为性能提升留下了空间。

IIN任务中的最优方法。 为了进行公平对比，我们使用两种地图表示（回合地图和场景特定地图）评估了先前IIN最优方法 [20], [22]的性能。*Mod-IIN* [20]将IIN任务分解为探索、目标实例重识别、目标定位和局部导航。该方法利用特征匹配在以自我为中心的视觉中重新识别目标实例，并将匹配的特征投影到地图上以定位目标。每个子任务都使用无需微调的现成组件解决。

IEVE [22]采用模块化架构，能够在探索、验证和利用动作之间动态切换。这种灵活性使智能体能够根据不同情况做出明智的决策。*Mod-IIN*和*IEVE*在IIN任务上都表现出了出色的性能。我们还在这些方法中实现了场景特定地图表示，以便与GaussNav进行公平对

Ablations	Success $\uparrow$	SPL $\uparrow$
GaussNav	0.725	0.578
GaussNav w.o. Classifier	0.375	0.291
GaussNav w.o. Match	0.444	0.353
GaussNav w.o. NVS	0.716	0.557
GaussNav w. SIFT	0.655	0.519
GaussNav w. GlueStick [84]	0.723	0.577
GaussNav w. GT Match	0.850	0.672
GaussNav w. GT Goal Localization	0.946	0.744

表 II

ABLATION STUDY OF GAUSSNAV. WE STUDY THE IMPACT OF CLASSIFIER, MATCH MODULE, NOVEL VIEW SYNTHESIS (NVS) AND DIFFERENT LOCAL FEATURE EXTRACTION AND MATCHING ALGORITHMS ON OUR GAUSSNAV. THE LAST TWO ROWS DESCRIBE THE PERFORMANCE OF OUR METHOD USING GROUND TRUTH MATCH RESULTS AND GOAL POSITION.

Method	Metric	Ch.	So.	Pl.	Bed	Tol.	TV
w.o. clf	Success $\uparrow$	0.821	0.873	0.798	0.914	0.936	0.829
	Time (s) $\downarrow$	30.9	24.8	33.2	18.1	24.6	31.9
w. clf	Success $\uparrow$	0.782	0.859	0.854	0.878	0.702	0.847
	Time (s) $\downarrow$	15.7	8.09	11.6	5.97	1.85	2.74

表 III

RESULTS OF MATCHING SUCCESS AND TIME TAKEN WITH OR WITHOUT CLASSIFIER.(ABBREVIATIONS: CLF = CLASSIFIER, CH. = CHAIR, SO. = SOFA, PL. = PLANT, TOL. = TOILET)

比。如Table I所示，GaussNav在所有方法中都表现出显著的性能优势。它在SPL指标上以0.231的巨大幅度超越了所有现有模型（最后两行）。这一结果表明，我们的语义高斯地图通过保留场景中物体的精细纹理细节，使智能体能够基于目标图像直接定位目标物体，而无需额外的验证步骤。

我们将这种卓越的性能归因于语义高斯这一新颖的地图表示，它使智能体无需额外的探索或验证即可直接定位目标，如Figure 6所示。与之前广泛使用的BEV地图表示不同，我们的语义高斯地图允许智能体通过语义标签输入查询高斯，并渲染物体实例的描述性图像。因此，与基于BEV地图的方法不同，我们的GaussNav不需要对潜在的物体候选进行显式验证。相反，它从众多可能性中选择最可能的候选，并直接向其导航。

Chair	Sofa	TV	Plant	Toilet	Bed
11	2	1	3	1	0

表 IV

NUMBER OF OBJECT INSTANCES ACROSS DIFFERENT CATEGORIES IN THE FIRST FLOOR OF SCENE CrMo8WxCyVb.

³<http://www.locobot.org/>

Metric	original	GT original	horizontal		vertical		GT horizontal		GT vertical	
	$n_v = 1$	$n_v = 1$	$n_v = 3$	$n_v = 5$	$n_v = 3$	$n_v = 5$	$n_v = 3$	$n_v = 5$	$n_v = 3$	$n_v = 5$
Success $\uparrow$	0.811	0.815	0.824	0.827	0.832	0.831	0.835	0.846	0.839	0.845
Time (s) $\downarrow$	11.5	11.7	32.1	53.9	33.1	54.2	32.7	54.1	33.4	55.1

表 V

RESULTS OF THE NUMBER OF RENDERED IMAGES n , DIFFERENT DIRECTIONS (VERTICAL OR HORIZONTAL) AND WHETHER TO USE GROUND TRUTH RENDERING'S IMPACT ON THE MATCHING SUCCESS.

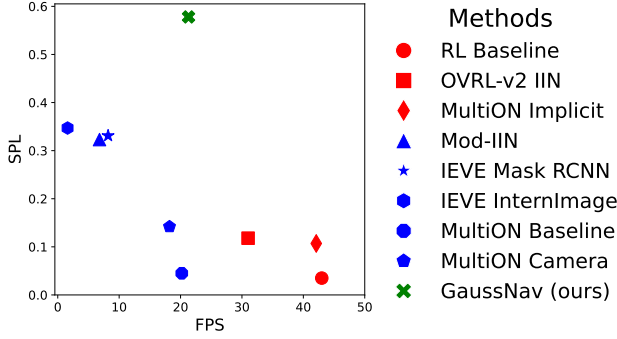


图 7. SPL和帧率分析。红色标记代表端到端方法，蓝色标记代表模块化方法。我们的GaussNav属于模块化方法，在模块化方法中实现了最高的帧率，同时也在所有IIN任务方法中取得了最高的SPL。

C. 消融实验

为了理解GaussNav各个模块的作用，我们进行了以下消融实验：

移除分类器的GaussNav。 在Figure 5中，我们将分类器的输出替换为随机生成的目标类别。我们观察到成功率下降到0.375，SPL下降到0.291（Table II第2行）。为了更好地评估分类器，我们设计了以下实验。我们将来自所有候选物体的训练视图定义为全集，将经过分类器过滤的训练视图定义为子集。给定目标图像和关键点匹配器，我们分别在每个集合中计算匹配关键点数量最多的视图。如果该视图包含目标物体，则视为成功；否则视为失败。我们还记录了整个局部特征匹配过程所花费的总时间，以评估分类器对效率的影响。这里我们不使用成功率和SPL等导航指标，因为这样可以消除路径规划错误等无关因素的影响。实验结果如Table III所示。可以看出，没有分类器时的总成功率略高于有分类器时（提高了0.039），但提升有限。然而，所花费的时间是有分类器时的2.5倍，效率显著降低。因此，考虑到性能和效率之间的权衡，我们选择将分类器作为GaussNav框架的一个组件。

移除匹配模块的GaussNav。 匹配模块旨在从同类别候选中区分出目标。没有匹配模块时，我们从这些候选中随机选择。成功率和SPL分别下降到0.444和0.353（Table II第3行）。这可归因于没有匹配模块时对候选实

Method	Ver. NVS	Hor. NVS	GT	Success $\uparrow$	SPL $\uparrow$
IEVE [22]	-	-	-	0.702	0.252
GaussNav (ours)	✓	×	×	0.713	0.259
GaussNav (ours)	×	✓	×	0.715	0.261
GaussNav (ours)	✓	✓	×	0.723	0.265
GaussNav (ours)	✓	✓	✓	0.747	0.289

表 VI

PERFORMANCE COMPARISON OF IEVE [22] AND GAUSSNAV. WE COMPARE GAUSSNAV WITH THE STATE-OF-THE-ART METHOD IEVE USING DIFFERENT MAP REPRESENTATIONS. (ABBREVIATIONS: VER. = VERTICAL, HOR. = HORIZONTAL)

例的随机选择。

使用SIFT或GlueStick的GaussNav。 为了评估不同的特征提取器和匹配算法对GaussNav的影响，我们替换了DISK [78]+LightGlue [79]组合。我们使用SIFT+FLANN和GlueStick [84]作为替代方案。第一种替换导致性能下降较大（Table II第4行），第二种仅显示出轻微下降（Table II第5行）。这些结果表明，不同的局部特征匹配算法在HM3D数据集上表现出不同的性能。

新视角合成(NVS)分析。 我们进行实验以评估渲染图像数量 n_v 和NVS上限的影响。具体而言，我们在水平和垂直方向上分别对 $n_v = 3$ 和 $n_v = 5$ 的NVS进行了消融研究，并评估了使用真实值(GT)渲染结果可达到的上限。为了避免导航中其他因素的影响，我们仅将通过关键点匹配检索到的图像包含目标物体视为成功；否则视为失败。具体的评估指标在移除分类器的GaussNav部分有详细说明。实验结果如Table V所示。总体而言，使用NVS有助于成功识别物体。然而，较大的 n_v 并不一定产生积极效果（见Table V，垂直 $n_v = 5$ 与垂直 $n_v = 3$ 对比）。这是因为我们的语义高斯地图对于特定物体可能只有少数几个观测视角，而传统3DGS中通常有几十个。因此，从新视角渲染时可能会出现孔洞等伪影，减少了特征数量，如Figure 8所示。尽管如此，对于真实值NVS，性能提升是显著的。

为了进一步研究无需预探索的NVS的影响，我们采用了与IEVE [22]相同的框架，唯一的区别在于地图表示。IEVE使用二维BEV地图，而GaussNav实现了语

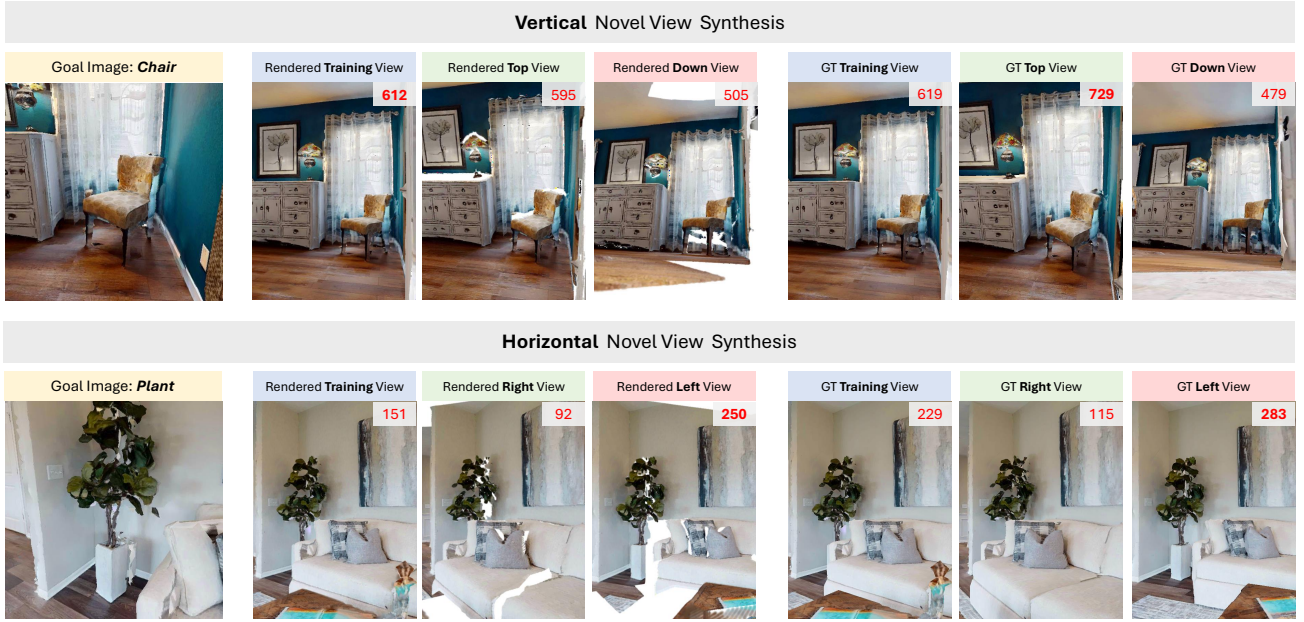


图 8. 新视角合成结果可视化。可视化结果包括 $\theta = \pm 15^\circ$ 的水平和垂直新视角合成结果。每张图像的右上角显示了与目标图像匹配的关键点数量。

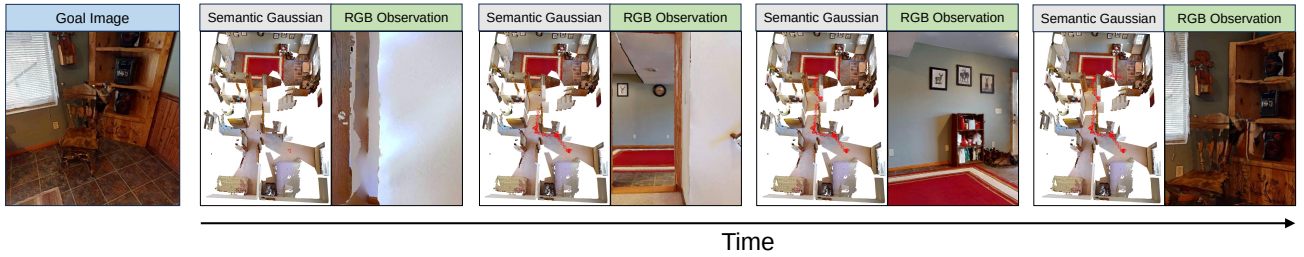


图 9. GaussNav智能体在Habitat仿真器中执行IIN任务的定性示例。当在环境中随机初始化时，智能体被提供一张描绘目标物体的目标图像。高斯导航直接在语义高斯地图中定位目标物体，并引导智能体向其移动。

义高斯地图。此外，GaussNav在每次观测中出现相似物体时都会执行NVS。结果如Table VI所示。我们的语义高斯地图在不依赖预探索的情况下增强了现有观测。虽然这种提升可能不如实例级物体定位能力那么显著，但Table VI最后一行的结果表明了未来工作的有希望方向。

D. 效率分析

我们从预处理和智能体-环境交互两个角度分析了我们的方法的时间效率。在我们的方法中，“预处理”指的是给定语义高斯地图后，匹配和定位目标物体的过程。后者则比较了我们的方法与各种其他方法的运行时帧率。

在地图中高效定位目标物体对我们的方法至关重要，我们利用渲染结果来匹配和定位目标物体。为了降低在每个可导航视点比较所有可能观测的计算成本，我们引入了语义高斯。该技术将物体实例按其对应的语义标签进行分组。这种优化通过将比较限制在每个实例

的几个描述性渲染图上，显著缩小了搜索空间。例如，在CrMo8WxCyVb场景中，可导航区域量化为 $54.03m^2$ 。该空间可离散化为约54个正方形，每个面积为 $1m^2$ 。智能体位于每个正方形内时，可以从12个不同的视角观察周围环境，覆盖完整的 360° ，每个视角跨度 30° 。因此，该单层的原始搜索空间将包含 $12 \times 54 = 648$ 个潜在观测。通过应用基于语义标签的分组策略，搜索空间被大幅缩小。我们统计了CrMo8WxCyVb场景中不同类别的物体实例数量，如Table IV所示。为了定位一把“椅子”——假设我们从三个独特的视角渲染每个物体实例——最终的搜索空间减少到仅 $3 \times 11 = 33$ 个观测。我们渲染单个实例的三个观测以确保渲染质量，这只有在新视角与训练视图大部分重叠时才能实现。这种优化带来了时间效率的显著提升。

我们还在单个Habitat环境中使用NVIDIA GeForce RTX 3090 GPU和10个CPU核心比较了不同方法的运行时帧率。如Figure 7所示，我们的方法保持了超过20 FPS的高效率，同时也在所有对比方法中取得了最高



图 10. 语义高斯可视化结果。

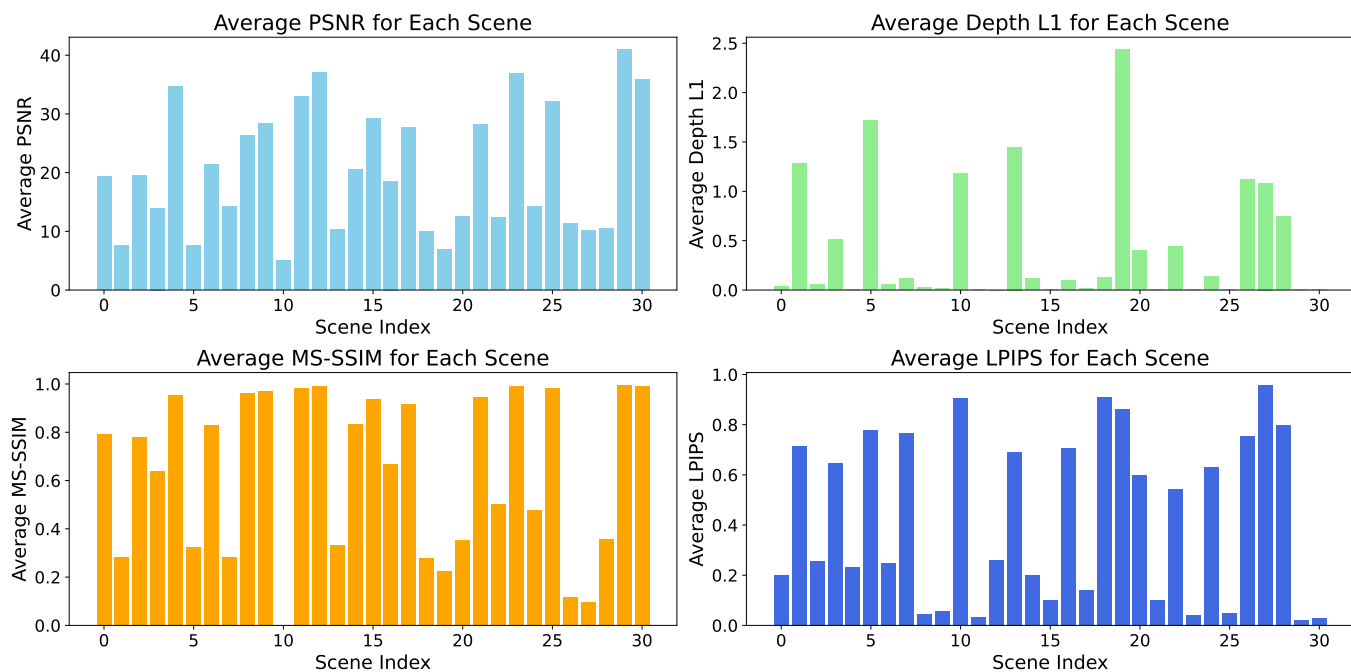


图 11. 我们的语义高斯构建方法在HM3D验证数据集上的渲染质量。x轴表示不同的场景索引及其对应的楼层高度。

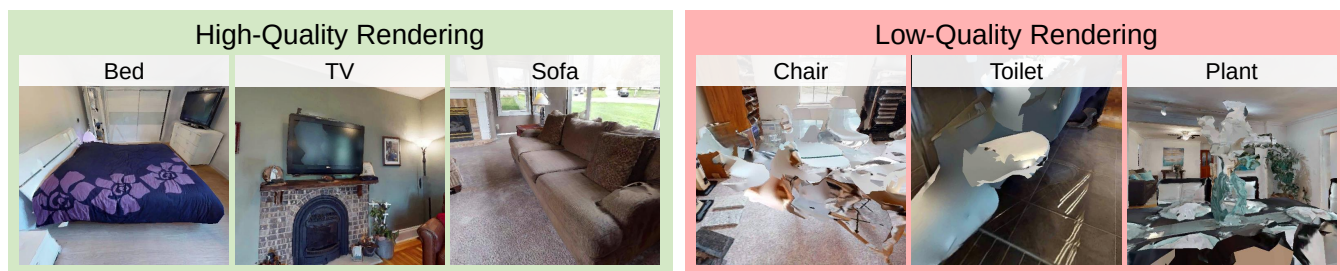


图 12. 使用Habitat仿真器 [16]从HM3D [14]场景数据集渲染的观测图像。

的SPL。为了实现更高的运行速度，GaussNav将语义高斯地图投影得到二维栅格地图。利用“预处理”阶段预测的目标位置，我们然后使用快速行进法(FMM)进行路径规划。我们的方法在速度上优于各种模块化方法的原因是，我们在导航过程中不依赖语义分割 [20], [22]、局部特征匹配 [20], [22]或切换模块 [22]等额外组件。这种简化使GaussNav能够更高效地运行。我们还提供了GaussNav导航至目标物体实例的定性示例，如Figure 9所示。

E. 错误分析

所提模型的性能仍远非完美。我们希望了解错误模式以便未来改进。我们的分析确定了两个错误来源：第一个是模型无法从实例渲染图中一致地匹配目标，第二个是目标定位不准确。为了量化这些错误来源的影响，我们使用真实值匹配模块和精确的目标定位对模型进行了评估。第一个意味着智能体可以从候选观测中正确识别目标，第二个意味着智能体直接获得真实值目标位置。配备真实值匹配模块的变体（Table II中的**GaussNav w. GT Match**）表明，成功率可以提高约0.127（Table II第5和第6行）。此外，当模型同时配备真实值匹配和目标定位时，记为**GaussNav w. GT Goal Localization**，我们观察到成功率从0.850提高到0.946，如Table II第6和第7行所示。第一个错误来源的改进可以通过开发更鲁棒的重识别算法来实现。至于第二个来源，更精确的定位策略可能会产生更好的结果。这些见解不仅突出了模型当前的不足，还为后续的改进工作指明了方向。

F. 高斯构建结果

如Figure 10所示，我们提供了语义高斯的可视化结果。这些示例证明了我们的语义高斯表示在各种场景中的有效性。通过展示更广泛的结果，我们旨在说明我们的方法在处理各种场景复杂度和物体组合方面的鲁棒性和适用性。

在Figure 11中，我们对语义高斯构建方法在HM3D验证数据集 [14]上产生的渲染质量进行了定量评估。为了符合IIN任务 [19]的约束，我们将HM3D中的每个场景划分为独立的楼层，并限制智能体在单个楼层内移动，因为IIN任务 [19]本身确保智能体的起始位置和目标位置在同一楼层。

为了定量分析Figure 11中的结果，我们观察到渲染结果呈现出两极分化的趋势。例如，在索引为29和30的

场景中，渲染图像达到了高达40的PSNR和接近零的深度渲染误差。然而，场景10的渲染性能不佳。我们假设不同场景之间渲染质量的这种两极分化可归因于仿真与现实之间的差异。这在Figure 12中很明显，其中使用Habitat仿真器从HM3D数据集渲染的一些图像显示出低保真度，特别是在高纹理环境中。在这种复杂环境中进行高质量重建是困难的，而使用次优渲染图作为三维环境重建的基础会进一步降低最终输出的质量。鉴于此，对于重建质量较差的场景，我们保持一致性，使用原始训练视图，而不是尝试渲染可能导致质量下降的新视图。

V. 结论

在本工作中，我们提出了一种用于视觉导航的模块化方法，即面向视觉导航的三维高斯泼溅技术(GaussNav)。现有的基于地图的方法主要专注于构建二维鸟瞰图(BEV)地图，这类地图无法表示场景的三维几何结构和精细特征。为此，我们提出了一种新颖的地图表示——语义高斯地图，它能够保留场景的三维几何结构、每个高斯对应的语义标签以及精细的纹理细节。利用这种新颖的地图表示，我们能够直接预测目标图像中所描绘物体的位置，从而将实例图像目标导航(IIN)任务转化为更易处理的点目标导航(PointGoal Navigation)任务。我们提出的框架取得了当前最优的性能，将路径长度加权成功率(SPL)从0.347显著提升至0.578。此外，我们分析了模型的错误模式，并量化了未来工作中在两个重要维度（匹配和物体定位）上的改进空间。

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GaussNav: Gaussian Splatting for Visual Navigation

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Abstract—In embodied vision, Instance ImageGoal Navigation (IIN) requires an agent to locate a specific object depicted in a goal image within an unexplored environment. The primary challenge of IIN arises from the need to recognize the target object across varying viewpoints while ignoring potential distractors. Existing map-based navigation methods typically use Bird’s Eye View (BEV) maps, which lack detailed texture representation of a scene. Consequently, while BEV maps are effective for semantic-level visual navigation, they are struggling for instance-level tasks. To this end, we propose a new framework for IIN, Gaussian Splatting for Visual Navigation (GaussNav), which constructs a novel map representation based on 3D Gaussian Splatting (3DGS). The GaussNav framework enables the agent to memorize both the geometry and semantic information of the scene, as well as retain the textural features of objects. By matching renderings of similar objects with the target, the agent can accurately identify, ground, and navigate to the specified object. Our GaussNav framework demonstrates a significant performance improvement, with Success weighted by Path Length (SPL) increasing from 0.347 to 0.578 on the challenging Habitat-Matterport 3D (HM3D) dataset. The source code is publicly available at the link: <https://github.com/XiaohanLei/GaussNav>.

Index Terms—Embodied Visual Navigation, 3D Gaussian Splatting.

I. INTRODUCTION

EMBODIED visual navigation is an emerging computer vision problem where an agent uses visual sensing to actively interact with the world and perform navigation tasks [1]–[12]. Recent years have witnessed substantial progress in embodied visual navigation, fueled by the availability of large-scale photo-realistic 3D scene datasets [13]–[15] and fast simulators for embodied navigation [16]–[18]. These advancements have enabled researchers to develop and test navigation algorithms in controlled environments that closely mimic real-world conditions.

In embodied visual navigation, one critical question is “how can we describe the goal target?”. One common setting for tasking an agent is to give it natural language instructions, such as “check if my laptop is on the chair”. However, this setting becomes confusing when there are multiple chairs in the house. To overcome this challenge, Krantz *et al.* [19]

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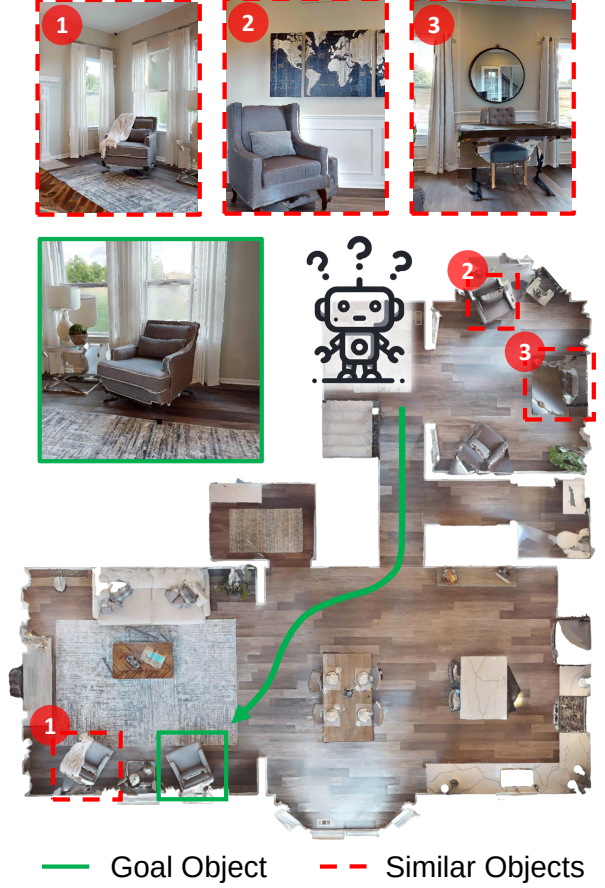


Fig. 1. Illustration of Instance ImageGoal Navigation (IIN), which requires agent to navigate to the object instance depicted in the goal image, while distinguishing it from other visually similar instances.

propose Instance ImageGoal Navigation (IIN) [20]–[22]. In IIN, an agent is presented with an image of a specific object, and its goal is to navigate to the specific object within the least time budget. The goal image is not expected to match the sensor specification or embodiment of the navigating agent, as described in Figure 1. To accomplish the task, the agent needs to distinguish the target object from different angle of views and ignore potential distractors. This is a challenging task as it involves semantic reasoning, geometry understanding and instance-aware matching.

To address the above issue, previous methods [20], [22]–[27] introduce 2D Semantic Bird-Eye-View (BEV) map to tackle this problem. These well-designed explicit map representations are memory-efficient, storing essential information such as 2D geometry and semantics, and can be directly utilized to calculate the agent’s subsequent actions. However, this simple 2D BEV map representation lacks the capacity to retain 3D geometrical information about the environment, rendering it ineffective for navigating cross-floor scenarios. Additionally,

BEV maps are unable to preserve the instance-aware features in a scene, which can be crucial for distinguishing between multiple objects of the same class or for tasks requiring fine-grained object interaction.

To avoid the limitation of BEV maps, we propose a new Gaussian Splatting Navigation framework, *i.e.*, GaussNav, for IIN task. Our GaussNav is inspired by the recent advancements in 3D Vision technologies, including Neural Radiance Fields (NeRF) [28] and 3D Gaussian Splatting (3DGS) [29]. These technologies have demonstrated substantial progress in novel view synthesis (NVS) and 3D scene understanding. Although these technologies were not initially developed for navigation tasks, they offer considerable potential for application in this domain. To this end, we develop the Semantic Gaussian map representation, which integrates the representation of geometry, semantics and instance-aware features, and can be directly used for visual navigation.

Our GaussNav framework consists of three stages, including Frontier Exploration, Semantic Gaussian Construction and Gaussian Navigation. First, the agent employs Frontier Exploration to collect observations of the unknown environment. Second, the collected observations are used to construct Semantic Gaussian. By leveraging semantic segmentation algorithms [30], [31], we assign semantic labels to each Gaussian. We then cluster Gaussians with their semantic labels and 3D positions, segmenting objects in the scene into different instances under various semantic categories. This representation is capable of preserving not only the 3D geometry of the scene and the semantic labels of each Gaussian, but also the texture details of the scene, thereby enabling NVS. Third, we render descriptive images for object instances, matching them with the goal image to effectively locate the target object. Upon determining the predicted goal object's position, we can efficiently transform our Semantic Gaussian into grid map and employ path planning algorithms to accomplish the navigation.

To the best of our knowledge, we are the first to introduce 3DGS [29] to embodied visual navigation. In this work, we unify the map representation of geometry, semantics and instance-aware features for visual navigation. Our framework can directly ground the target object with a single goal image input and guide the agent towards it without any additional exploration or verification [22]. Our framework designs are beneficial for effective and efficient visual navigation. We evaluate our method's on both efficacy and efficiency and establish new state-of-the-art records on the challenging Habitat-Matterport 3D dataset (HM3D) [14].

II. RELATED WORK

We briefly discuss related work on differentiable rendering, followed by a broad overview of embodied visual navigation, and finally, we focus on the work most relevant to us: Instance ImageGoal Navigation (IIN).

Differentiable Rendering. To achieve photo-realistic scene capture, differentiable volumetric rendering has gained prominence with the introduction of Neural Radiance Fields (NeRF) [28]. NeRF utilizes a single Multi-Layer Perceptron (MLP) to represent a scene, performing volume rendering by marching

along pixel rays and querying the MLP for opacity and color. Due to the inherent differentiability of volume rendering, the MLP representation is optimized to minimize rendering loss using multi-view information, resulting in high-quality novel view synthesis (NVS). The primary limitation of NeRF is its slow training speed. Recent advancements have addressed this issue by incorporating explicit volume structures, such as multi-resolution voxel grids [32]–[34] and hash functions [35], to enhance performance.

In contrast to NeRF, 3DGS [29] employs differentiable rasterization. Unlike ray marching, which iterates along pixel rays, 3DGS iterates over the primitives to be rasterized, similar to conventional graphics rasterization. By leveraging the natural sparsity of a 3D scene, 3DGS achieves an expressive representation capable of capturing high-fidelity 3D scenes while offering significantly faster rendering. Comprehensive review of the developments in 3D Gaussian Splatting [36] highlights the method's versatility and applications across various domains. Leveraging these advantages, a growing body of research begins to explore various innovations, including deformable or dynamic Gaussians [37]–[39], advancements in mesh extraction and physics simulation [40]–[42], as well as applications in Simultaneous Localization and Mapping (SLAM) [43]–[45]. This surge of interest in the capabilities of 3DGS leads us to consider its potential for embodied visual navigation. Given that the Gaussian representation inherently encapsulates explicit scene geometry and the parameters required for rendering, we posit that 3DGS can enhance decision-making in map-based visual navigation. The explicit nature of the Gaussian representation provides a rich, condensed form of environmental data, which can be effectively utilized to inform and guide autonomous agents in navigation.

Embodied Visual Navigation. Embodied Visual navigation includes several topics: ObjectGoal Navigation (ObjectNav), Multi ObjectGoal Navigation (MultiON), ImageGoal Navigation (ImageNav), and Instance ImageGoal Navigation (IIN). ObjectNav [24], [46]–[49] requires an agent to navigate to any instance of a specified object category within the environment. MultiON [50]–[53], on the other hand, tasks the agent to sequentially navigate to a series of objects. ImageNav [26], [54]–[61] involves navigating to the camera pose from which a target image is captured. In contrast, IIN [20]–[22] requires navigating to the specific instance captured by the camera in the target image. Collectively, these navigation tasks span a spectrum from semantic-level navigation in ObjectNav and MultiON to fine-grained instance-level navigation in ImageNav and IIN, comprehensively capturing the problem space of embodied visual navigation.

Numerous approaches to solving embodied visual navigation utilize deep reinforcement learning (DRL) to develop end-to-end policies that map egocentric vision to action [26], [46], [56], [62]. However, acquiring skills related to visual scene understanding, semantic exploration, and long-term memory are challenging in an end-to-end framework. Consequently, these methods often incorporate a combination of careful reward shaping [54], pre-training routines [56], and advanced memory modules [59], [63], [64]. In contrast to end-to-end DRL, alternative approaches decompose the problem into sub-

tasks that can be optimized in a supervised manner. These sub-tasks include graph prediction via topological SLAM [58], graph-based distance learning [65], [66], and camera pose estimation for last-mile navigation [61]. Chaplot *et al.* [24] decompose the embodied visual navigation task into exploration, object detection, and local navigation. Building on this, CLIP on Wheels (CoW) [67] utilizes a similar decomposition strategy, focusing on exploration and object localization to handle an open-set object vocabulary. Modular approaches also show promise for effective simulation-to-reality transfer (Sim2Real). Gervet *et al.* [68] conduct a Sim2Real transfer of both modular and end-to-end systems, demonstrating that modularity effectively mitigates the visual Sim2Real gap that impairs the performance of end-to-end policies.

Modular approaches typically represent the environment using a map and utilize this map to acquire the knowledge necessary for navigation within that environment. Bird’s-Eye View (BEV) maps [20], [22], [24], [69] are a form of metric map that project the entire scene from an overhead perspective, representing the occupancy status, exploration state, and semantic categories of specific areas. This map representation encodes information about each region into a grid, achieving accurate spatial awareness. In contrast, topological maps [57], [58], [60] focus more on describing the spatial relationships between different areas in a scene. This approach allows agents to navigate based on the connectivity of spaces rather than relying solely on geometric coordinates. To represent a scene more meticulously, 3D-aware maps have been proposed [49], [70]–[72]. Incorporating 3D consistency can enhance perception in navigation. Building upon 3D maps, we further advance by adopting 3D Gaussian Splatting (3DGS) [29] as a novel map representation. This enables the map to synthesize appearance views of specific object instances, which we have demonstrated to be effective in the Instance ImageGoal Navigation task.

Instance ImageGoal Navigation. The IIN task introduces distinct challenges compared to the ImageGoal Navigation. First, in IIN, the goal image must depict a specific object instance. In contrast, ImageGoal Navigation may use randomly captured photos that could include insignificant elements, such as large white walls. Second, the camera parameters used to capture the goal image do not necessarily match those of the agent’s camera. Therefore, to succeed in IIN, an agent must be adept at identifying the target object among numerous candidates of the same class with goal object and recognizing it from various viewpoints. To address the above challenges, Krantz *et al.* [20] develop a general pipeline for aligning the same object from different angle of views. Bono *et al.* [21] present an end-to-end approach in the IIN task, while Lei *et al.* [22] propose a method that mimics human behaviour for verifying objects at a distance. Existing methods focus partly on designing sophisticated modules [20], [22] and partly on pre-training on pretext tasks [21]. Our approach differs in that we concentrate on designing a new map representation. Through this novel form of map, we can better establish the connection between target description and target locations, thereby facilitating visual navigation.

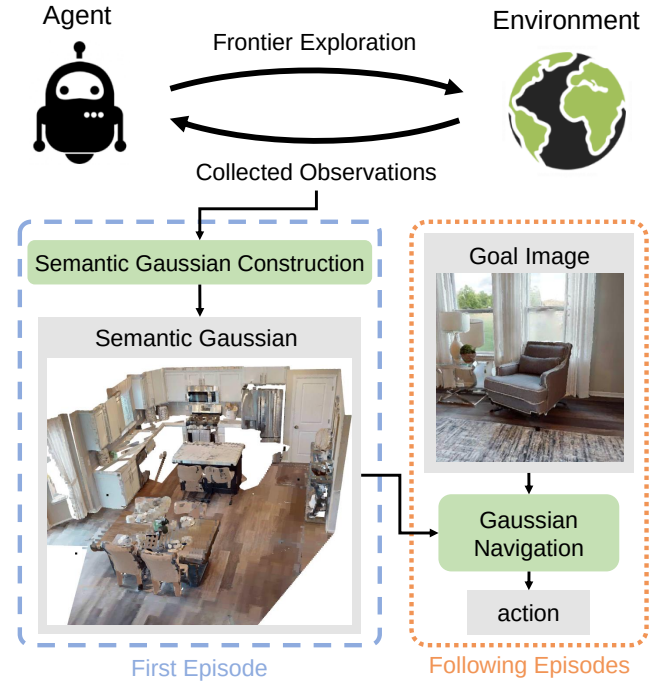


Fig. 2. Framework Overview. In the first episode of a scene, the agent uses frontier exploration to gather observations of the unknown environment, constructing a Semantic Gaussian. In subsequent episodes, the pre-constructed Semantic Gaussian is utilized by Gaussian Navigation to ground the goal object and guide the agent towards it.

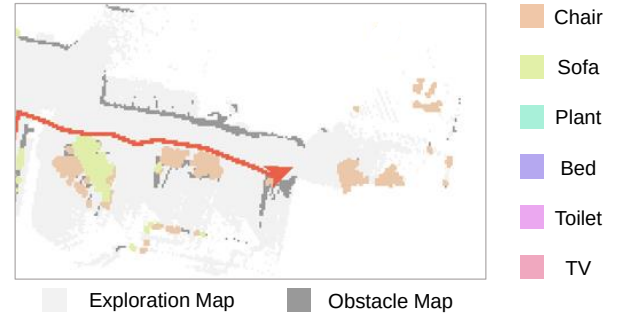


Fig. 3. Exploration Map and Obstacle Map.

III. METHODS

A. Overview

In the IIN task, at the start of a new episode e , the agent is given a goal image I_g that features a specified object instance O_g . The agent’s goal is to navigate to the referred instance O_g within the environment. At each timestep t , the agent acquires observations which include an RGB image I_t , a depth image D_t , and sensor pose reading P_t . Utilizing this information, the agent must decide upon and execute an action a_t . When calling stop action, the episode is considered successful only if the agent is within a certain range of the goal object.

To accomplish the IIN task, we propose a modular framework called Gaussian Splatting for Visual Navigation (Gauss-Nav), as depicted in Figure 2. In a new environment, the Instance ImageGoal Navigation in our proposed framework consists of three stages: Frontier Exploration, Semantic Gaussian Construction, as depicted in Figure 4, and Gaussian Navigation, illustrated in Figure 5. Initially, during the first

episode within an unknown environment, the agent employs frontier exploration to explore the environment and collect observations. We then use our proposed Semantic Gaussian to reconstruct the scene. Subsequently, in the following episodes, the agent leverages the Semantic Gaussian to ground the object instance o depicted in the goal image and navigate to it. This process effectively transforms the IIN task into a more manageable PointGoal Navigation task.

B. Frontier Exploration

In the first episode of an unexplored environment, the agent simultaneously maintains two types of maps, an exploration map and an obstacle map, as illustrated in Figure 3. The exploration map delineates the regions of the environment that have been explored, while the obstacle map marks the obstacles in the scene. By detecting the contours of the exploration map and excluding areas in the obstacle map, the agent sets the closest frontier point as a waypoint to facilitate exploration. The frontier-based exploration strategy is a well-established approach in robotics and autonomous navigation [73], [74]. It involves identifying the boundaries between the explored and unexplored regions of the environment, known as frontiers. The agent then selects the nearest accessible frontier point as its next target for exploration. This decision is based on the distance to the frontier, the navigability of the path, and the potential information gain from exploring that frontier [75]. By iteratively exploring the nearest frontiers, the agent efficiently covers the entire environment while avoiding obstacles and previously explored areas. Through the application of this frontier-based exploration strategy, the agent collects observations of the entire environment.

C. Semantic Gaussian Construction

Our Semantic Gaussian is represented by a group of Gaussians, with each Gaussian characterized by a minimal set of nine parameters: a triplet for the RGB color vector $\mathbf{c}$, a triplet delineating the centroid $\boldsymbol{\mu} \in \mathbb{R}^3$, a scalar representing the radius r , a scalar quantifying the opacity $o \in [0, 1]$, and a scalar representing the category label l . Different from original 3DGS [29], we simplify the representation of Gaussian by using only view-independent color and constraining the Gaussian to be isotropic. This simplification enhances computational efficiency while reducing memory requirements. For a comprehensive understanding of 3DGS [29], we highly recommend readers consulting the original paper [29]. Our Semantic Gaussian Construction can be described as an iterative process comprising two alternating steps: Gaussian Densification and Semantic Gaussian Updating, as depicted in Figure 4. Gaussian Densification initializes new Gaussians in the Semantic Gaussian at each new incoming frame, while Semantic Gaussian Updating refines the parameters of each Gaussian through Differentiable Rendering.

Differentiable Rendering. 3DGS [29] renders an RGB image as follows: given a collection of 3D Gaussians and camera pose, first sort all Gaussians from front to back. RGB images can then be efficiently rendered by alpha-compositing the

splatted 2D projection of each Gaussian in order in pixel space. The rendered color of pixel $\mathbf{p} = (u, v)$ can be written as:

$$\hat{I}(\mathbf{p}) = \sum_{i=1}^n \mathbf{c}_i f_i(\mathbf{p}) \prod_{j=1}^{i-1} (1 - f_j(\mathbf{p})), \quad (1)$$

where $f_i(\mathbf{p})$ is computed as follows:

$$f(\mathbf{x}) = o \exp\left(-\frac{\|\mathbf{x} - \boldsymbol{\mu}\|^2}{2r^2}\right). \quad (2)$$

The $\boldsymbol{\mu}$ and r are the splatted 2D Gaussians in pixel-space:

$$\boldsymbol{\mu}^{2D} = K \frac{E_t \boldsymbol{\mu}}{d}, \quad r^{2D} = \frac{fr}{d}, \quad \text{where } d = (E_t \boldsymbol{\mu})_z. \quad (3)$$

Here, K is the camera's intrinsic matrix, E_t embodies the extrinsic matrix that encodes the camera's rotation and translation at time t , f denotes the known focal length, and d is the depth of the i^{th} Gaussian relative to the camera's coordinate frame.

Render. Different from 3DGS [29], we differentially render depth and silhouette image, which determines the visibility and will be used for the next Gaussian Densification and Updating. The depth D and silhouette image S at pixel $\mathbf{p}$ is rendered as follows:

$$\hat{D}(\mathbf{p}) = \sum_{i=1}^n d_i f_i(\mathbf{p}) \prod_{j=1}^{i-1} (1 - f_j(\mathbf{p})), \quad (4)$$

$$\hat{S}(\mathbf{p}) = \sum_{i=1}^n f_i(\mathbf{p}) \prod_{j=1}^{i-1} (1 - f_j(\mathbf{p})). \quad (5)$$

Above these, we also render the semantic segmentation results as follows:

$$\hat{C}(\mathbf{p}) = \sum_{i=1}^n l_i f_i(\mathbf{p}) \prod_{j=1}^{i-1} (1 - f_j(\mathbf{p})). \quad (6)$$

Semantic Segmentation. As depicted in Figure 4, the agent's RGB observation I_t is segmented into C_t using Mask-RCNN [30]. The segmented result will be used for initializing new Gaussians in Gaussian Densification and supervising the Gaussians' parameters in Semantic Gaussian Updating.

Gaussian Densification. Gaussian Densification is performed by comparing the rendered results at position P_t using Gaussians from $t-1$ with the ground truth. This process adds new Gaussians where the previous Gaussians fail to represent the scene in the new observations. Following Keetha *et al.* [43], we add new Gaussians based on a densification mask to determine which pixels should be densified:

$$M(\mathbf{p}) = \left(\hat{S}(\mathbf{p}) < 0.5\right) + \left(D(\mathbf{p}) < \hat{D}(\mathbf{p})\right) \left(L_1(\hat{D}(\mathbf{p})) > 50\text{MDE}\right), \quad (7)$$

where the first term represents where the Semantic Gaussian is not adequately dense, and the second term indicates where the ground-truth depth is in front of the predicted depth and the depth error is greater than 50 times the median depth error (MDE).

Semantic Gaussian Updating. After densifying current Semantic Gaussian, we update the parameters of Gaussians

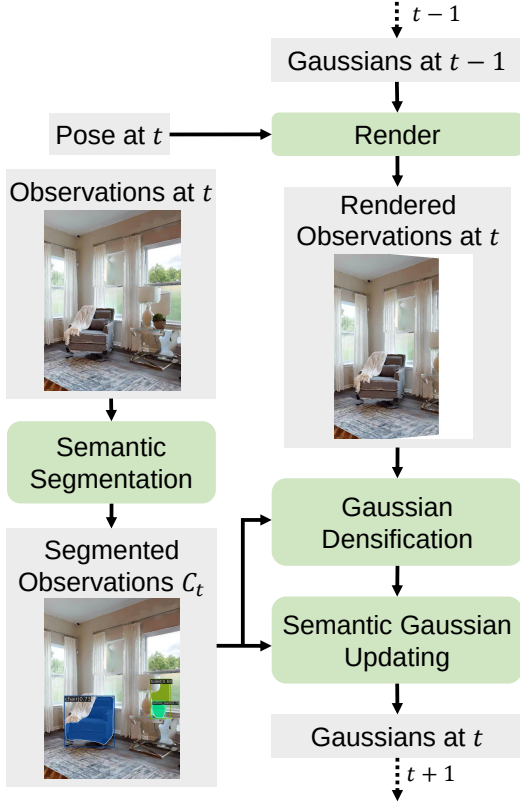


Fig. 4. An illustration of Semantic Gaussian Construction. At timestep t , the pipeline updates the Gaussians from $t - 1$ through densification and updating, which involves a comparison between the rendered RGB and depth images against the current input training views. Concurrently, semantic labels are assigned to the densified Gaussians using the segmented images. Finally, the Gaussians are refined through differentiable rendering.

given poses and observations. This is done by differentiable-rendering and gradient-based-optimization, which is equivalent to the “classic” problem of fitting a radiance field to images with known poses. Specifically, we update the parameters of Gaussians by minimizing the RGB, depth and segmentation errors.

D. Gaussian Navigation

To navigate using a constructed Semantic Gaussian, we propose Gaussian Navigation, as illustrated in Figure 5. We first classify the goal image I_g to predict the semantic label $\hat{l}_g$ of the current target, such as ‘chair’ in Figure 5. This semantic label is then used to query the relevant Gaussians. For each relevant object instance of the same class, we render descriptive images. These renderings are compared with the goal image to ground the goal object, predicting its position $\hat{P}_g$. Finally, the Path Planning module generates a feasible path and determines the agent’s action.

Classifier. In IIN task, the agent receives an image depicting the target object I_g . However, comparing this image with renderings from navigable points within the entire scene becomes exceedingly time-inefficient due to the vastness of the search space. Consequently, with our Semantic Gaussian G_s , we only need to search for instances of the object category corresponding to the goal object. Therefore, we first classify

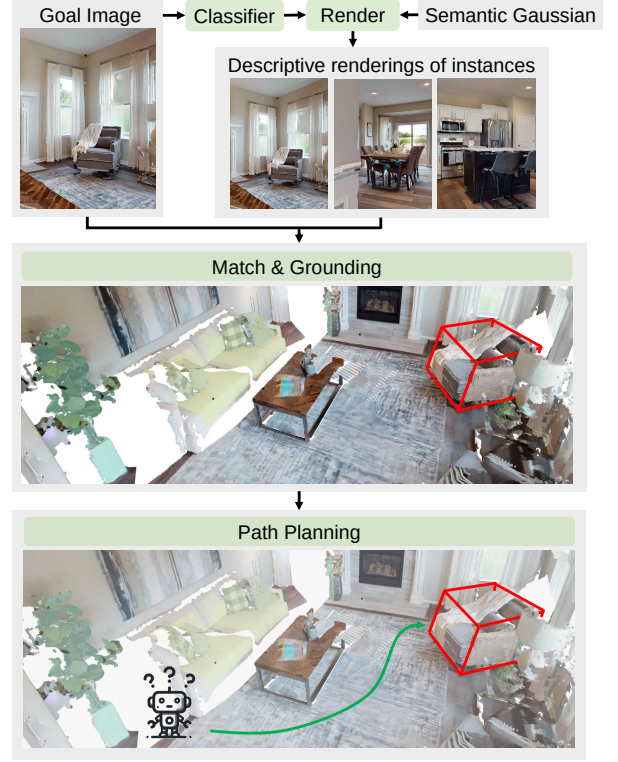


Fig. 5. An illustration of Gaussian Navigation. Our approach begins with the classification of the goal image using pre-constructed Semantic Gaussian. Upon determining the predicted class, we generate descriptive images around instances belonging to that class. These images are then matched with the target object to identify and ground the goal instance. Utilizing the map and the established goal, the agent employs path planning to compute the sequence of actions.

the goal image I_g into target category label $\hat{l}_g$. We use the goal images on the train split of HM3D-SEM [14] to finetune the image classification model, *i.e.*, ResNet50 [76], pretrained on ImageNet [77].

Match & Grounding. With the predicted target object’s label $\hat{l}_g$, we identify all candidate objects that share the same class label. For each candidate instance, we generate a set of descriptive images by rendering the object from multiple viewpoints to capture its features. Specifically, for one training view containing possible candidate objects, we augment it to n_v views by novel view synthesis (NVS). We denote the transformation from the camera to the world coordinate system of the training view as $c2w$, and record the translation of the potential target object in the camera frame as $\mathbf{t}_c^o$. We define the rotation matrix from the object to the world frame using the *forward*, *right*, and *up* vectors. The *forward* direction is the vector from the training view to the potential target object, namely $\mathbf{t}_c^o$. The *up* direction is the vector $[0, -1, 0]$, and the *right* vector is orthogonal to both the *forward* and *up* vectors, forming a right-handed coordinate system with the *right*, *up*, and *forward* vectors. The translation vector from the object to the world frame is

$$\mathbf{t}_w^o = c2w \times \mathbf{t}_c^o. \quad (8)$$

Together, the rotation matrix and translation vector constitute the rigid transformation matrix $o2w$ from the object to the

world frame.

We define the rotation matrices around the y -axis and x -axis as $\mathbf{R}_y(\theta)$ and $\mathbf{R}_x(\theta)$, respectively, representing new viewpoints formed by rotating around the object by an angle θ in the horizontal and vertical directions. Thus, the transformation from the camera to the world frame for new viewpoints in the horizontal direction is

$$\mathbf{c2w}_h(\theta) = \mathbf{o2w} \times \mathbf{R}_y(\theta) \times \mathbf{w2o} \times \mathbf{c2w}, \quad (9)$$

and in the vertical direction:

$$\mathbf{c2w}_v(\theta) = \mathbf{o2w} \times \mathbf{R}_x(\theta) \times \mathbf{w2o} \times \mathbf{c2w}. \quad (10)$$

In experiments, when $n_v = 1$, we do not perform NVS; when $n_v = 3$, we perform NVS at $\theta = \pm 15^\circ$ (both horizontal and vertical); and when $n_v = 5$, we use $\theta = \pm 15^\circ, \pm 30^\circ$.

After NVS, the original training views are augmented. The augmented rendering set for the i -th object instance is denoted as S_i . Let n denote the observed instances of the same class as target object, then the universal set of S can be formulated as:

$$S = \{S_1, S_2, \dots, S_n\} \quad \text{for } i = 1, 2, \dots, n. \quad (11)$$

To distinguish the target object from these candidates, we can formulate the question as:

$$i_{\max} = \arg \max \left\{ \max_{s \in S_1} \Omega(s), \max_{s \in S_2} \Omega(s), \dots, \max_{s \in S_n} \Omega(s) \right\} \quad (12)$$

for $i = 1, 2, \dots, n$,

where $\Omega(\cdot)$ is defined as the matched number of keypoints between renderings and goal image I_g . Specifically, for the rendering $s \in S_i$ of the i -th object instance and the goal image I_g , we extract the pixel-wise (x, y) coordinates of the keypoints and their associated feature descriptors V_t using DISK [78]. That is:

$$(K_t, V_t) = \text{DISK}(s), \quad (K_g, V_g) = \text{DISK}(I_g). \quad (13)$$

Subsequently, the matched pairs $(\hat{K}_t, \hat{K}_g)$ are computed using LightGlue [79]. The feature matching process is formulated as follows:

$$(\hat{K}_t, \hat{K}_g) = \text{LightGlue}((K_t, V_t), (K_g, V_g)). \quad (14)$$

Thus, the number of matched points, *i.e.*, the length of $\hat{K}_t$ or $\hat{K}_g$, is denoted as Ω . The candidate object whose rendered images yield the highest number of matched keypoints is then selected as the target object, as shown in Equation (12).

When the object instance is selected, we ground the object in the Semantic Gaussian. Due to the presence of outliers, which are caused by errors in semantic segmentation, we perform clustering on the instances on the map. Specifically, we use Density-based Spatial Clustering of Applications with Noise (DBSCAN) ¹, which groups together points that are closely packed together, marking as outliers points that lie alone in low-density regions. With the precise selected object instance's location, we can easily transform the IIN task to a PointGoal task.

Path Planning. The Semantic Gaussian is not designed for path planning. We first convert the Semantic Gaussian into a point cloud, whereby each Gaussian is reduced to a single point in the point cloud representation. The point cloud is then voxelized into 3D voxels M_{3D} , then the 3D voxels M_{3D} are projected to 2D BEV grids M_{2D} . Here, we use the 2D projection of Semantic Gaussian rather than the 2D geometric map used in pre-exploration to maintain a consistent representation throughout the pipeline. Our Semantic Gaussian initializes Gaussians using point cloud derived from depth image and does not prune any Gaussian in the optimization stage. Thus, 2D projection of Semantic Gaussian or 2D geometric map are fundamentally equivalent.

Given 2D BEV grid map M_{2D} , along with the agent's starting position and the goal's location, we can efficiently generate a shortest distance field using FMM. Each point within this field encapsulates the minimal distance necessary to traverse from the starting point to the goal. We extract a relevant subset of this distance field that falls within the agent's operational range. Subsequently, a waypoint is chosen from this subset to ensure it avoids any intersections with obstacles while adhering to a local minimum in the distance field. With the selected waypoint, the agent can readily calculate an action based on the angle and distance from its current state. The agent iterates the aforementioned process to generate a sequence of actions, continuing until the destination is reached.

IV. EXPERIMENTS

This section first details our experiment setup, followed by a comparative analysis with state-of-the-art approaches. Subsequently, we present an ablation study to evaluate the efficacy of individual components in our proposed methods. Finally, we assess the computational efficiency of our approach, examine error cases, and evaluate the rendering quality.

A. Experiment Setup

Datasets. We use Habitat-Matterport 3D dataset (HM3D) [14] in the Habitat [16] for our experiments. HM3D consists of scenes which are 3D reconstructions of real-world environments with semantic annotations. These scenes are split into three distinct subsets for training, validation, and testing, consisting of 145/36/35 scenes, respectively. We follow the task setting of Instance ImageGoal Navigation (IIN) proposed by Krantz *et al.* [19]. The episode dataset has been partitioned into three subsets for training, validation, testing, comprising 7,056K/1K/1K episodes respectively. The object depicted by the goal image belongs to the following six categories: {"chair", "couch", "plant", "bed", "toilet", "television"}. On the validation subset, a total of 795 unique object instances have been observed.

Embodiment. We adopt the embodiment parameters from the Hello Robot Stretch platform ². The simulated agent is modeled as a rigid-body cylinder with zero turning radius, a height of 1.41m, and a radius of 0.17m. A forward-facing RGB-D camera is affixed at a height of 1.31 m. At each

¹<https://scikit-learn.org/stable/modules/generated/sklearn.cluster.DBSCAN.html>

²<https://hello-robot.com/stretch-2>

Method	Success $\uparrow$	SPL $\uparrow$
RL Baseline [19]	0.083	0.035
OVRL-v2 ImageNav [55]	0.006	0.002
OVRL-v2 IIN [55]	0.248	0.118
FGPrompt [80]	0.099	0.028
MultiON Baseline [51]	0.066	0.045
MultiON Implicit [53]	0.143	0.107
MultiON Camera [50]	0.186	0.142
Mod-IIN [20]	0.561	0.233
IEVE Mask RCNN [22]	0.684	0.241
IEVE InternImage [22]	0.702	0.252
Mod-IIN [20] (Scene Map)	0.563	0.323
IEVE Mask RCNN [22] (Scene Map)	0.683	0.331
IEVE InternImage [22] (Scene Map)	0.705	0.347
GaussNav (ours)	0.725	0.578

TABLE I

PERFORMANCE COMPARISON OF OUR GAUSSNAV WITH PREVIOUS STATE-OF-THE-ART METHODS ON THE HM3D [14] DATASETS ACROSS TWO DIFFERENT METRICS: SUCCESS AND SPL [81]. THE TABLE IS DIVIDED INTO FOUR SECTIONS. THE FIRST SECTION PRESENTS THE RESULTS OF END-TO-END METHODS. THE SECOND SECTION SHOWS THE TRANSFER PERFORMANCE OF MULTIION-RELATED METHODS ON THE IIN TASK. THE THIRD SECTION INCLUDES THE STATE-OF-THE-ART METHODS ON THE IIN TASK. FINALLY, THE FOURTH SECTION AIMS TO PROVIDE A FAIR COMPARISON WITH GAUSSNAV BY REPLACING THE EPISODIC MAP USED IN THESE METHODS FROM THE THIRD SECTION WITH A SCENE-SPECIFIC MAP, ALLOWING THE AGENT TO RETAIN THE MAP FROM THE PREVIOUS EPISODE.

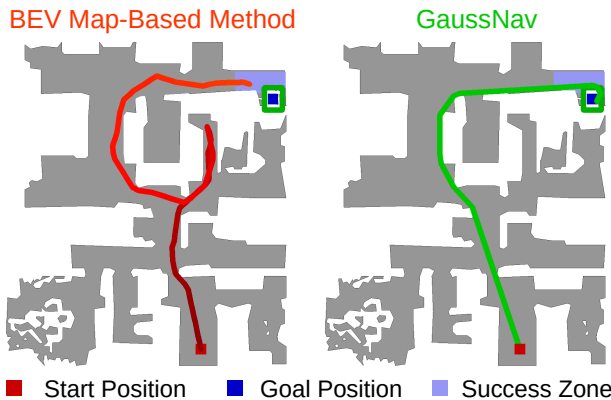


Fig. 6. Trajectory Analysis. Our Semantic Gaussian map representation can allow agent to directly ground target object from a single goal image, thereby facilitating efficient visual navigation.

timestep t , the agent’s observation consists of an egocentric RGB image, depth image, goal image and sensor pose reading. Camera specifications, such as mounting height, look-at angle, and field of view (FOV), differ between the agent’s and the goal’s cameras. Specifically, the agent’s camera resolution is 640×480 , whereas the goal’s camera has a resolution of 512×512 with unfixed height and FOV parameters.

Action Space. We use a discrete action space for navigation, comprising four actions: {STOP, FORWARD, TURN_RIGHT, TURN_LEFT}. The STOP action terminates the current episode, while the FORWARD action advances the agent by 25 cm. Rotational actions occur in place: TURN_RIGHT induces a 25-degree clockwise rotation and TURN_LEFT a 25-degree counter-clockwise rotation.

Evaluation Metrics. Following Krantz *et al.* [19], we evaluate our model with both success and efficiency. We report

Success Rate (Success), Success rate weighted by normalized inverse Path Length (SPL). An episode is deemed successful (Success = 1) if the agent invokes the STOP action within a 1.0m Euclidean distance from the goal object. SPL is an efficiency measure defined in [81], is given by:

$$\text{SPL} = \frac{1}{N} \sum_{i=1}^N S_i \cdot \frac{l_i}{\max(p_i, l_i)}, \quad (15)$$

where N is the total number of episodes, S_i is a binary success indicator for episode i , l_i is the shortest path distance from the start position to the goal, and p_i is the path length actually traversed by the agent. A higher SPL value indicates more efficient navigation.

B. Comparison to the State-of-the-art Methods

We evaluate our proposed model against various baselines and previous state-of-the-art work, as presented in Table I. Unlike conventional IIN methods that generate a map for each episode, GaussNav constructs a map across episodes in a scene. The first block lists results from end-to-end baselines, and the second block includes methods designed for the MultiON task but implemented here for the IIN task. The third block of Table I presents the performance of original state-of-the-art IIN methods, while the fourth block reports their performance when adapted to use scene map.

End-to-end Baselines. We evaluate the performance of two end-to-end methods on the IIN task. *RL Baseline* built a network that observes agent RGB ($\mathcal{V}_{RGB}$), agent depth ($\mathcal{V}_D$), the goal image ($\mathcal{V}_G$), GPS coordinates (x, z), and heading (θ). Visual observations are encoded with separate ResNet-18 [82] encoders and GPS and heading are encoded with 32-dimensional linear layers. Then, these features are concatenated and encoded with a 2-layer LSTM. Finally, an action $a(t)$ is sampled from a categorical distribution. The network was trained from scratch using Proximal Policy Optimization (PPO) [83].

OVRL-v2, proposed by Yadav *et al.* [55], introduced self-supervised pretraining for visual encoders in ImageGoal Navigation. Originally, OVRL-v2 was trained for the ImageGoal Navigation task. Direct application of OVRL-v2 to IIN task without fine-tuning yields suboptimal performance, as indicated by a Success of 0.006 (row 2 in Table I). This reduced efficacy can be attributed to several factors: the transition between scene datasets from Gibson [13] to HM3D [14], differences in robot embodiment from Locobot³ to Stretch, and a shift in the nature of goal destinations from image sources to image subjects. Fine-tuning OVRL-v2 specifically for IIN task on the HM3D dataset significantly improves outcomes, resulting in a Success of 0.248 (row 3 in Table I).

State-of-the-art Methods in MultiON. The MultiON task shares many similarities with the Scene-specific Map representation we employ in evaluating GaussNav. We re-implement several state-of-the-art methods [50], [51], [53] from the MultiON task on the IIN benchmark. It is worth noting that we directly input the semantic category of the target object to the

³<http://www.locobot.org/>

Ablations	Success $\uparrow$	SPL $\uparrow$
GaussNav	0.725	0.578
GaussNav w.o. Classifier	0.375	0.291
GaussNav w.o. Match	0.444	0.353
GaussNav w.o. NVS	0.716	0.557
GaussNav w. SIFT	0.655	0.519
GaussNav w. GlueStick [84]	0.723	0.577
GaussNav w. GT Match	0.850	0.672
GaussNav w. GT Goal Localization	0.946	0.744

TABLE II

ABLATION STUDY OF GAUSSNAV. WE STUDY THE IMPACT OF CLASSIFIER, MATCH MODULE, NOVEL VIEW SYNTHESIS (NVS) AND DIFFERENT LOCAL FEATURE EXTRACTION AND MATCHING ALGORITHMS ON OUR GAUSSNAV. THE LAST TWO ROWS DESCRIBE THE PERFORMANCE OF OUR METHOD USING GROUND TRUTH MATCH RESULTS AND GOAL POSITION.

Method	Metric	Ch.	So.	Pl.	Bed	Tol.	TV
w.o. clf	Success $\uparrow$	0.821	0.873	0.798	0.914	0.936	0.829
	Time (s) $\downarrow$	30.9	24.8	33.2	18.1	24.6	31.9
w. clf	Success $\uparrow$	0.782	0.859	0.854	0.878	0.702	0.847
	Time (s) $\downarrow$	15.7	8.09	11.6	5.97	1.85	2.74

TABLE III

RESULTS OF MATCHING SUCCESS AND TIME TAKEN WITH OR WITHOUT CLASSIFIER. (ABBREVIATIONS: CLF = CLASSIFIER, CH. = CHAIR, SO. = SOFA, PL. = PLANT, TOL. = TOILET)

aforementioned methods. As the navigation requirements of the IIN task are at the instance level rather than the semantic level, this discrepancy in task formulation leaves room for performance improvement.

State-of-the-art Methods in IIN. For fair comparison, We evaluate the performance of previous state-of-the-art methods [20], [22] on the IIN task using two types of map representations: episodic map and scene-specific map. *Mod-IIN* [20] decomposes the IIN task into exploration, goal instance re-identification, goal localization, and local navigation. This method utilizes feature matching to re-identify the goal instance within the egocentric vision and projects the matched features onto a map to localize the goal. Each sub-task is addressed using off-the-shelf components that do not require any fine-tuning.

IEVE [22] employs a modular architecture that dynamically switches between exploration, verification, and exploitation actions. This flexibility empowers the agent to make informed decisions tailored to varying circumstances. *Mod-IIN* and *IEVE* both demonstrate exceptional performance on the IIN task. We also implement the scene-specific map representation in these methods for a fair comparison with GaussNav. As shown in Table I, GaussNav demonstrates a significant performance advantage over all methods. It significantly surpasses all existing models in terms of SPL by a huge margin of 0.231 (last 2 rows in Table I). This result indicates that our Semantic Gaussian map, by preserving the intricate texture details of objects in the scene, enables the agent to directly locate the target object based on the goal image without the need for additional verification.

We attribute the superior performance to the novel map representation of Semantic Gaussian, allowing agent to directly ground goal target without additional exploration or verification, as evidenced by Figure 6. Unlike previous widely-used BEV map representation, our Semantic Gaussian can

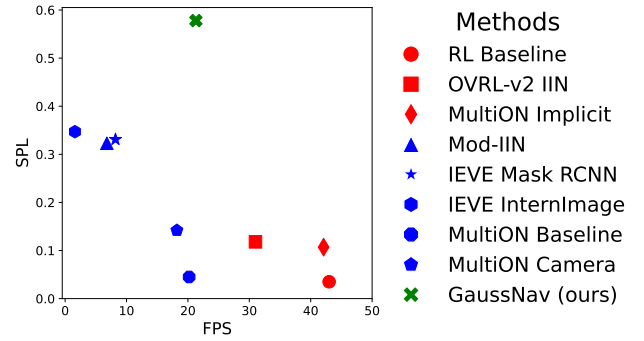


Fig. 7. SPL and FPS Analysis. Red markers represent end-to-end methods, while blue markers indicate modular approaches. Our GaussNav belongs to the modular approach, achieving the highest frame rate among modular methods while attaining the highest SPL across all approaches in IIN task.

Chair	Sofa	TV	Plant	Toilet	Bed
11	2	1	3	1	0

TABLE IV

NUMBER OF OBJECT INSTANCES ACROSS DIFFERENT CATEGORIES IN THE FIRST FLOOR OF SCENE CrM08WxCyVB.

allow agent to query Gaussian through semantic label input and render descriptive images of an object instance. Therefore, our GaussNav does not require explicit verification of potential object candidates, unlike BEV map-based approaches. Instead, it selects the most probable candidate from a multitude of possibilities and navigates towards it directly.

C. Ablation Study

To understand the modules of our GaussNav, we consider the following ablations:

GaussNav w.o. Classifier. In Figure 5, we replace the Classifier’s output with a random generated target category. We observe that the Success drops to 0.375 and the SPL decreases to 0.291 (row 2 in Table II). To better evaluate the classifier, we design the following experiment. We define the training views from the full space of candidate objects as the complete set, and the training views filtered by the classifier as a subset. Given the goal image and the keypoint matcher, we compute, in each set respectively, the view with the largest number of matching keypoints. If this view contains the goal object, we consider it a success; otherwise, a failure. We also record the total time spent in the entire local feature matching process to assess the impact of the classifier on efficiency. Here, we do not use the navigation metrics such as Success and SPL because this allows us to eliminate the influence of irrelevant factors, such as path planning errors. Our experimental results are shown in Table III. As can be seen, the total success without the classifier is slightly higher than that with the classifier (an increase of 0.039), but the improvement is limited. However, the time taken is 2.5 times that with the classifier, making it significantly less efficient. Therefore, considering the trade-off between performance and efficiency, we choose to include the classifier as a component of GaussNav framework.

GaussNav w.o. Match. The Match module is designed to distinguish the target from candidates with the same class. Without the Match module, we randomly select from these

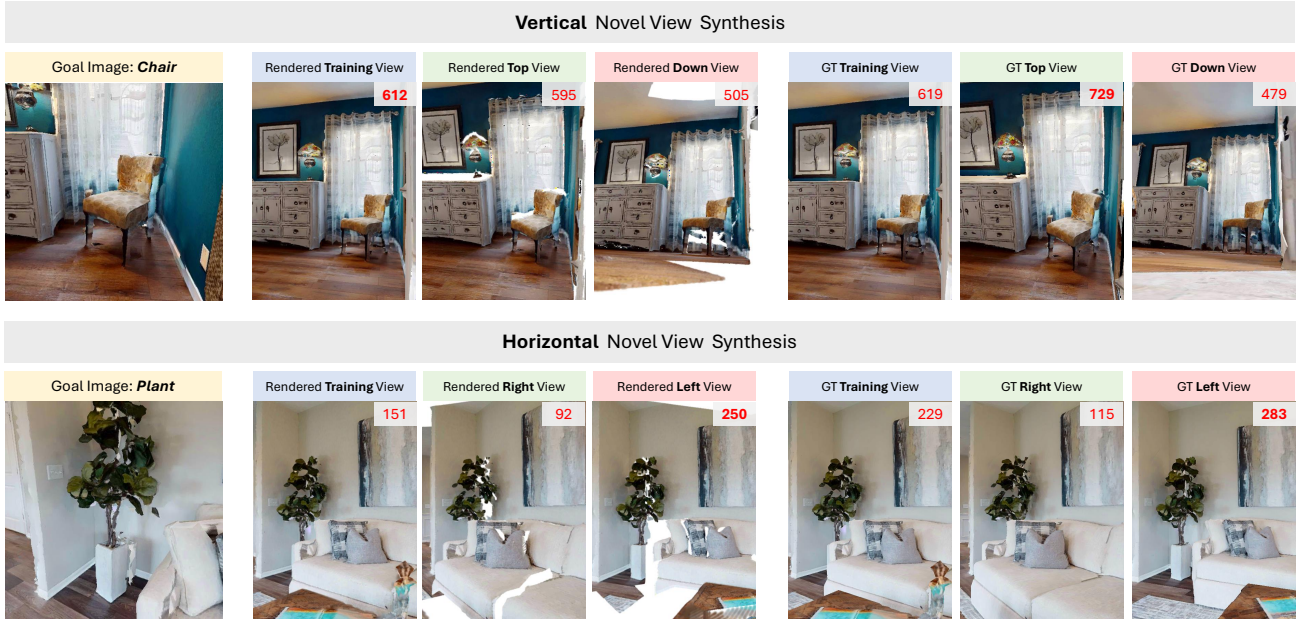


Fig. 8. Visualization of novel view synthesis results. The visualization results include horizontal and vertical novel view synthesis results with $\theta = \pm 15^\circ$. The upper right corner of each image shows the number of matched keypoints with the goal image.

Metric	original	GT original	horizontal		vertical		GT horizontal		GT vertical	
	$n_v = 1$	$n_v = 1$	$n_v = 3$	$n_v = 5$	$n_v = 3$	$n_v = 5$	$n_v = 3$	$n_v = 5$	$n_v = 3$	$n_v = 5$
Success $\uparrow$	0.811	0.815	0.824	0.827	0.832	0.831	0.835	0.846	0.839	0.845
Time (s) $\downarrow$	11.5	11.7	32.1	53.9	33.1	54.2	32.7	54.1	33.4	55.1

TABLE V

RESULTS OF THE NUMBER OF RENDERED IMAGES n , DIFFERENT DIRECTIONS (VERTICAL OR HORIZONTAL) AND WHETHER TO USE GROUND TRUTH RENDERING'S IMPACT ON THE MATCHING SUCCESS.

candidates. The Success and SPL falls to 0.444 and 0.353 (row 3 in Table II). This can be attributed to the random selection of candidate instances without Match module.

GaussNav w. SIFT or GlueStick [84]. To evaluate the impact of different extractors and matching algorithms on our GaussNav, we replace the combination of DISK [78] + LightGlue [79]. We employ both SIFT + FLANN and GlueStick [84] as alternatives. The first replacement represents a greater decrease, (row 4 in Table II) and the second only displays a slight decrease (row 5 in Table II). These results demonstrate the varying performance of different local feature matching algorithms on the HM3D dataset.

NVS Analysis. We conduct experiments to evaluate the impact of the number of rendered images n_v and the upper bound of NVS. Specifically, we perform ablation studies on NVS with $n_v = 3$ and $n_v = 5$ in both horizontal and vertical directions, and also evaluate the upper bound achievable using GT rendered results. To avoid the influence of other factors in navigation, we consider it a success only if the image retrieved through keypoint matching contains the target object; otherwise, it is considered a failure. The specific evaluation metrics are detailed in **GaussNav w.o. Classifier.** The experimental results are presented in Table V. Overall, utilizing NVS is beneficial for successfully recognizing objects. However, a large n_v does not necessarily yield positive effects (see Table V, vertical $n_v = 5$ vs. vertical $n_v = 3$). This is because our Semantic Gaussian map may have only a few observational viewpoints for a particular object, unlike the dozens available

Method	Ver. NVS	Hor. NVS	GT	Success $\uparrow$	SPL $\uparrow$
IEVE [22]	-	-	-	0.702	0.252
GaussNav (ours)	$\checkmark$	$\times$	$\times$	0.713	0.259
GaussNav (ours)	$\times$	$\checkmark$	$\times$	0.715	0.261
GaussNav (ours)	$\checkmark$	$\checkmark$	$\times$	0.723	0.265
GaussNav (ours)	$\checkmark$	$\checkmark$	$\checkmark$	0.747	0.289

TABLE VI

PERFORMANCE COMPARISON OF IEVE [22] AND GAUSSNAV. WE COMPARE GAUSSNAV WITH THE STATE-OF-THE-ART METHOD IEVE USING DIFFERENT MAP REPRESENTATIONS. (ABBREVIATIONS: VER. = VERTICAL, HOR. = HORIZONTAL)

in traditional 3DGS. Therefore, when rendering from novel viewpoints, artifacts such as holes may occur, reducing the number of features, as visualized in Figure 8. Nevertheless, for GT NVS, the improvement is significant.

To further study the impact of NVS without pre-exploration, we adopt the same framework as IEVE [22], with the only difference being the map representation. IEVE uses 2D BEV map while GaussNav implements Semantic Gaussian map. Additionally, GaussNav performs NVS whenever a similar object appears in each observation. The results are presented in Table VI. Our Semantic Gaussian map enhances the existing observations without relying on pre-exploration. While the improvement may not be as prominent as the ability of instance-level object localization, the result of last row in Table VI indicates promising direction for future work.

D. Efficiency Analysis

We analyze the temporal efficiency of our approach from two perspectives: preprocessing and agent-environment inter-

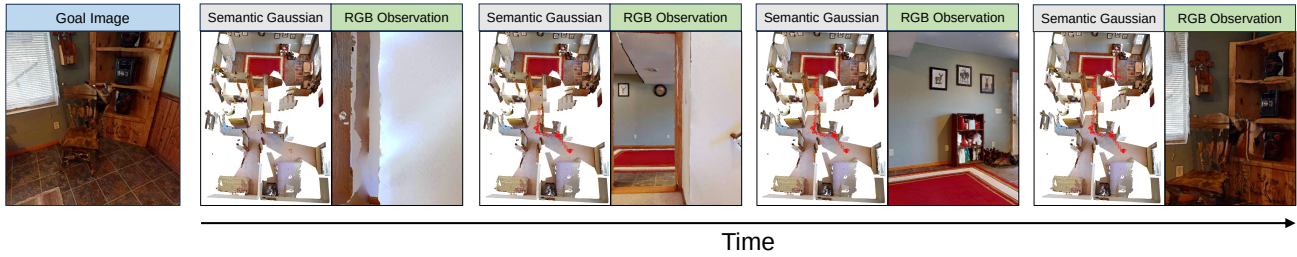


Fig. 9. Qualitative example of our GaussNav agent performing IIN task in the Habitat simulator. When randomly initialized in the environment, the agent is given a goal image depicted a target object. Gaussian Navigation directly grounds the target object in the Semantic Gaussian and guide the agent towards it.



Fig. 10. Semantic Gaussian visualization results.

action. In the context of our method, “preprocessing” refers to the process of matching and grounding target objects given the Semantic Gaussian. The latter perspective compares the runtime frame rate of our method with various other approaches.

Efficiently locating the target object within the map is crucial for our method, which leverages renderings to match and ground the goal object. To reduce the computational cost of comparing all possible observations at each navigable viewpoint, we introduce Semantic Gaussians. This technique groups object instances under their corresponding semantic labels. This optimization significantly reduces the search space by limiting comparisons to several descriptive renderings of each instance. For example, in the scene `CrMo8WxCyVb`, the navigable area is quantified to be $54.03m^2$. This space can be discretized into approximately 54 squares, each with an area of $1m^2$. Positioned within each square, the agent can observe its surroundings from 12 distinct viewing angles, covering a full 360 deg with each angle spanning 30 deg. Thus, the original search space for this single floor would consist of $12 \times 54 = 648$ potential observations. By applying our grouping strategy based on semantic labeling, the search space is considerably narrowed. We count the number of different categories of object instances within the `CrMo8WxCyVb` scene, as demonstrated in Table IV. To locate a “chair” — assuming we render each object instance from three unique viewpoints — the resulting search space is reduced to merely $3 \times 11 = 33$ observations. We render three observations of a single instance to ensure rendering quality, which is achievable only when the new viewpoint largely overlaps with the training views. This optimization yields a significant improvement in

time efficiency.

We also compare the runtime frame rates of different methods in a single Habitat environment using an NVIDIA GeForce RTX 3090 GPU and 10 CPU cores. As shown in Figure 7, our method maintains a high efficiency of over 20 FPS while achieving the highest SPL among the compared approaches. To achieve higher runtime speed, our GaussNav projects the Semantic Gaussian to obtain a 2D grid map. By utilizing the predicted target location from the “preprocessing”, we then employ Fast Marching Method (FMM) for path planning. The reason why our method outperforms various Modular Methods in terms of speed is that we do not rely on additional modules such as semantic segmentation [20], [22], local feature matching [20], [22], or switch module [22] during the navigation. This simplification enables GaussNav to operate more efficiently. We also provide a qualitative example of our GaussNav navigating to the target object instance, as depicted in Figure 9.

E. Error Analysis

The performance of the proposed model is still far from perfect. We would like to understand the error modes for future improvements. Our analysis identifies two sources of error: the first being the model’s inability to consistently match the target from instance renderings, and the second, inaccuracies in goal localization. To quantify the impact of these error sources, we conduct an evaluation of our model using a ground truth Match module and an accurate goal localization. The first one means agent can correctly recognize the target from candidate observations, and the second suggests agent is directly provided with the ground truth goal position.

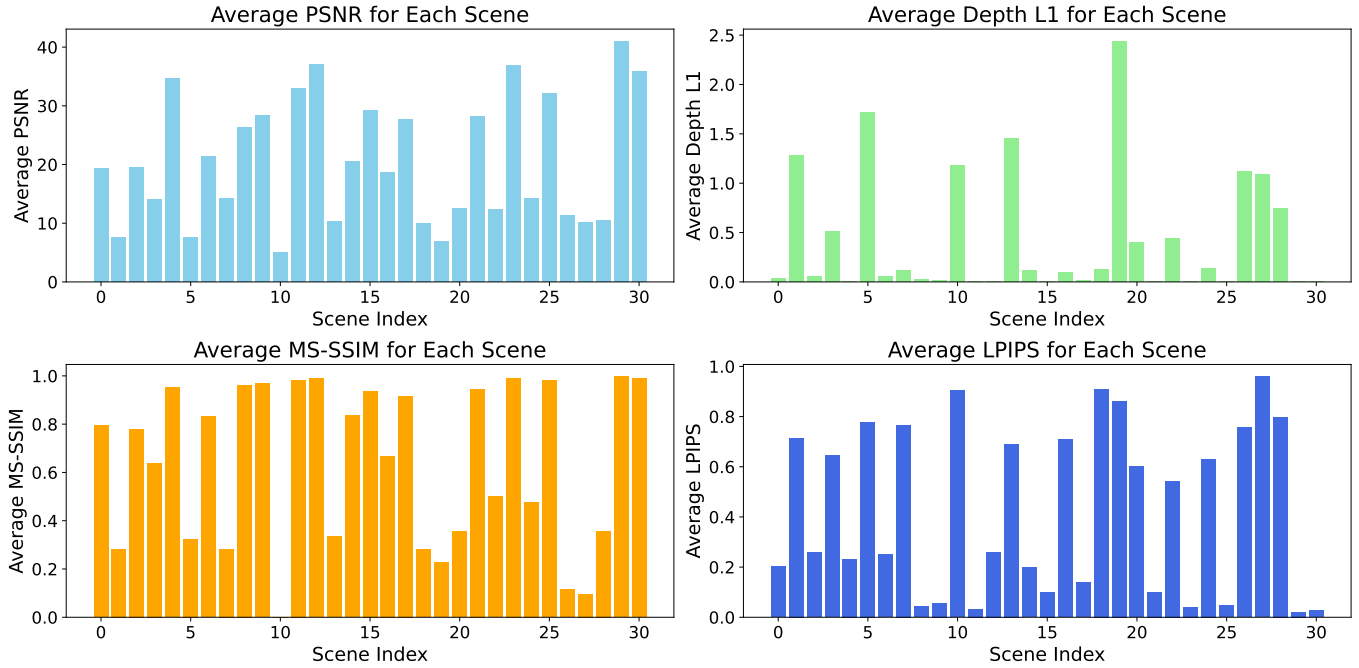


Fig. 11. Rendering quality of our Semantic Gaussian Construction results on the HM3D validation dataset. The x-axis indicates different scene indices with the corresponding floor height.

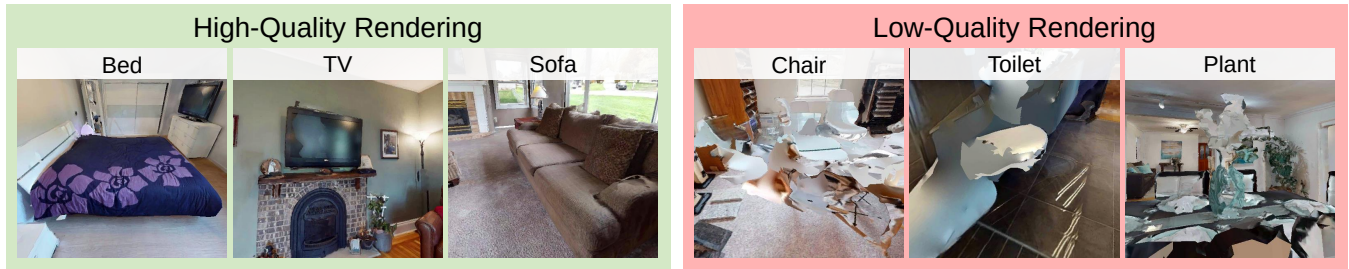


Fig. 12. Observations rendered from HM3D [14] scene dataset using the Habitat [16] simulator.

The variant equipped with a ground truth Match module (**GaussNav w. GT Match** in Table II) shows that Success can be enhanced by approximately 0.127 (rows 5 and 6 in Table II). Furthermore, when the model is augmented with both ground truth Match and Goal Localization, denoted as **GaussNav w. GT Goal Localization**, we observe an increase in Success from 0.850 to 0.946, as indicated in rows 6 and 7 in Table II. Improvements in the first error source may be achievable through the development of a more robust re-identification algorithm. As for the second source, a more precise Grounding strategy could yield better results. These insights not only highlight the model’s current shortcomings but also chart a course for subsequent refinement efforts.

F. Gaussian Construction Results

As illustrated in Figure 10, we provide visualization results of our Semantic Gaussian. These examples demonstrate the effectiveness of our Semantic Gaussian representation across a diverse range of scenarios. By presenting a more extensive collection of results, we aim to showcase the robustness and applicability of our approach in handling various scene complexities and object compositions.

In Figure 11, we present a quantitative evaluation of the rendering quality produced by our Semantic Gaussian Construction method on the HM3D validation dataset [14]. To align with the constraints of IIN task [19], we divide each scene within HM3D into separate floors and restrict the agent’s movement to within a single floor, as the IIN task [19] inherently ensures that both the agent’s starting location and the target’s position are on the same floor.

To quantitatively analyze the results in Figure 11, we observe that the rendering results exhibit a bifurcated trend. For instance, in scenes with indices 29 and 30, the rendered images achieve a high PSNR of up to 40 and a near-zero depth rendering error. However, the rendering performance for scene 10 is suboptimal. We hypothesize that this polarized rendering quality across different scenes can be attributed to the discrepancy between the simulation and reality. This is evident in Figure 12, where some renderings from the HM3D dataset using Habitat simulator exhibit low fidelity, particularly in highly textured environments. High-quality reconstruction in such intricate settings is difficult, and utilizing suboptimal renderings as a basis for 3D environment reconstruction can further degrade the quality of the final output. In light of

this, for scenes that are poorly reconstructed, we maintain consistency by using the original training views, rather than attempting to render novel views which would likely result in a diminished quality.

V. CONCLUSION

In this work, we introduce a modular approach for visual navigation, *i.e.*, Gaussian Splatting for Visual Navigation (GaussNav). Previous map-based methods largely focus on building 2D BEV map, which cannot represent the 3D geometry and detailed features in a scene. To this end, we propose a novel map representation, Semantic Gaussian, which is capable of preserving the scene's 3D geometry, semantic labels associated with each Gaussian, and intricate texture details. Leveraging this novel representation of map, we directly predict the position of target object depicted in the goal image, thereby transforming IIN into a more tractable PointGoal Navigation task. Our proposed framework achieves state-of-the-art performance, significantly enhancing SPL from 0.347 to 0.578. Furthermore, we analyze the error modes for our model and quantify the scope for improvement along two important dimensions (match and object grounding) in the future work.

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